

PSI

Center for Scientific Computing,
Theory and Data

Koopmans functionals in practice

minimisation, screening coefficients, automation, and more...

Edward Linscott

ICTP–MARVEL College, 27 May 2026

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

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Differences to semi-local functionals:

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Differences to semi-local functionals:

- different flavours

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Differences to semi-local functionals:

- different flavours
- orbital-density dependence

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Differences to semi-local functionals:

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- orbital-density dependence
- screening

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Flavours

Flavours of Koopmans functionals

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One degree of freedom: what should be the gradient of this linear term?

Flavours of Koopmans functionals



$$E_{\text{KI}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df \right)$$

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You might also see...

- “pKIPZ” = KIPZ Hamiltonian evaluated on the KI solution
- “K” = an earlier iteration based off half-filling rather than integer endpoints (no longer used)

Orbital-density-dependence

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$$-\int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df = E_{\text{Hxc}}[\rho] + E_{\text{Hxc}}[\rho - \rho_i] + f_i(-E_{\text{Hxc}}[\rho - \rho_i] + E_{\text{Hxc}}[\rho - \rho_i + n_i])$$

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For filled orbitals with KI:

$$v_i^{\text{KI}}/\alpha_i = -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}')$$

Orbital-density-dependence

Initialise with MLWFs, then (optionally) solve with CG minimisation:

G. Borghi *et al.* *Phys. Rev. B* **91**, 155112 (2015)

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For more details see

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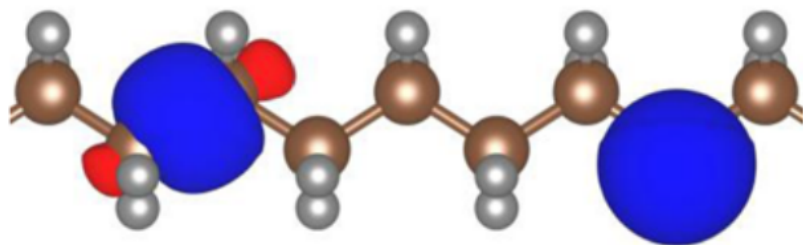
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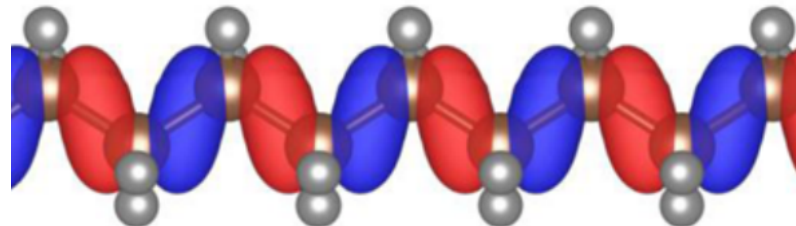
For more details see

Gives rise to a set of **minimising orbitals** (localised/variational); diagonalising at the minimum gives rise to **diagonalising orbitals** (delocalised/canonical)

Orbital-density-dependence



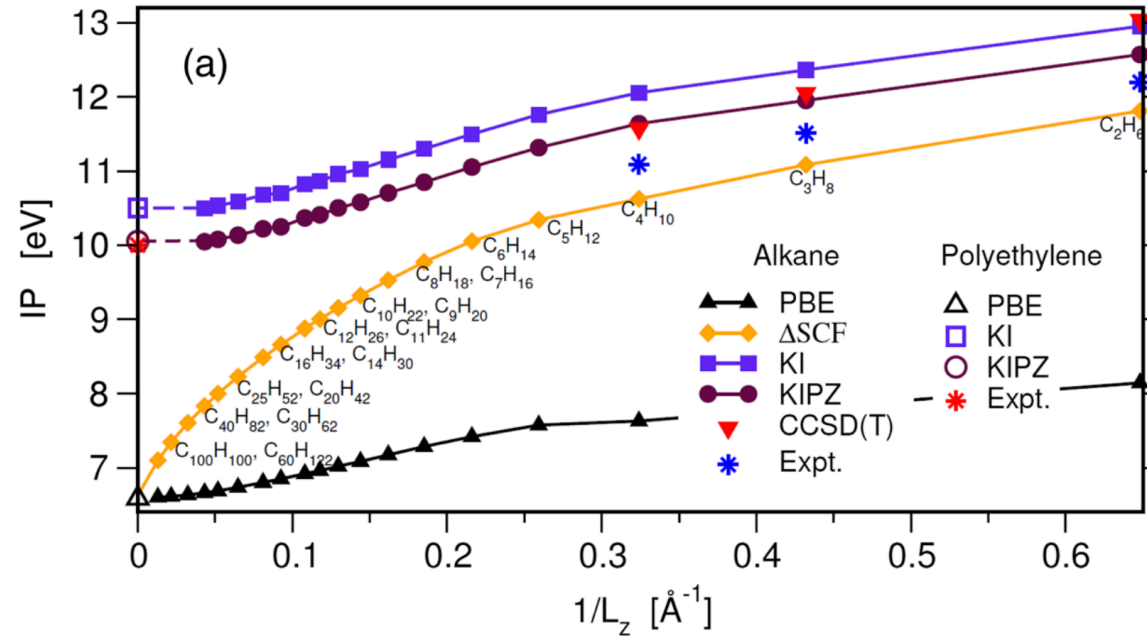
variational



canonical

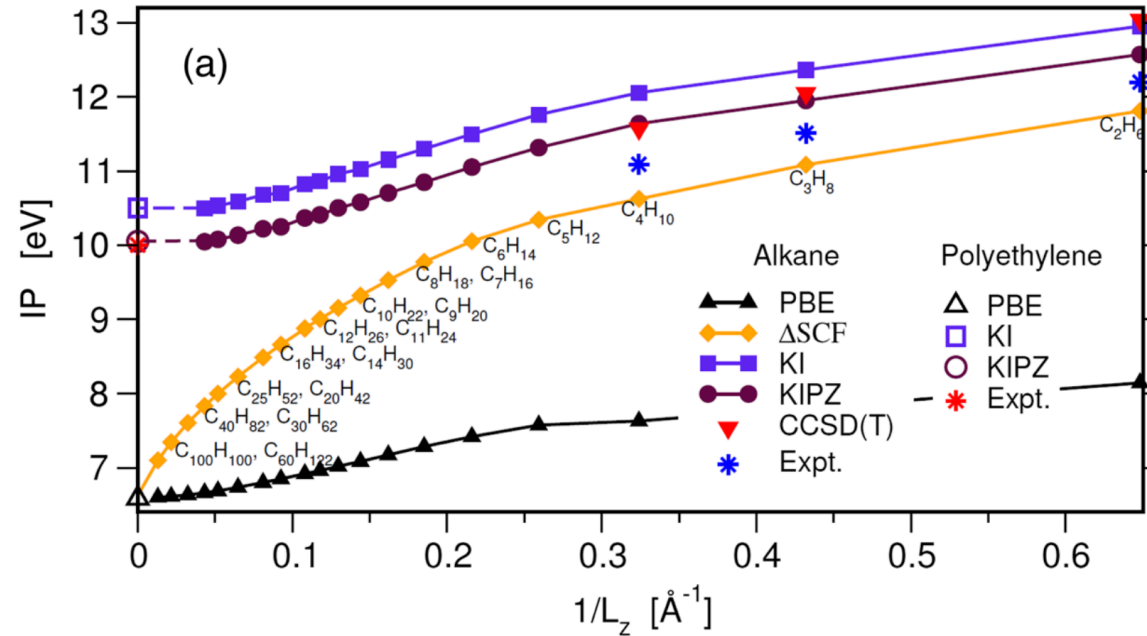
N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

Importance of localisation



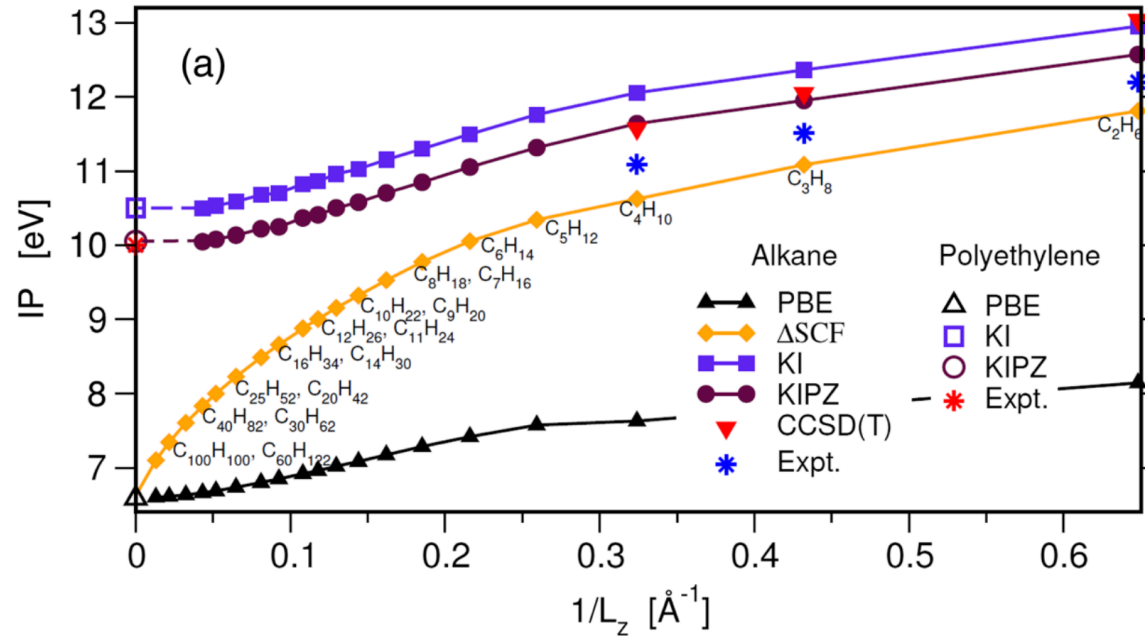
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Importance of localisation



In the bulk limit for one cell $\Delta E = E(N - \delta N) - E(N)$

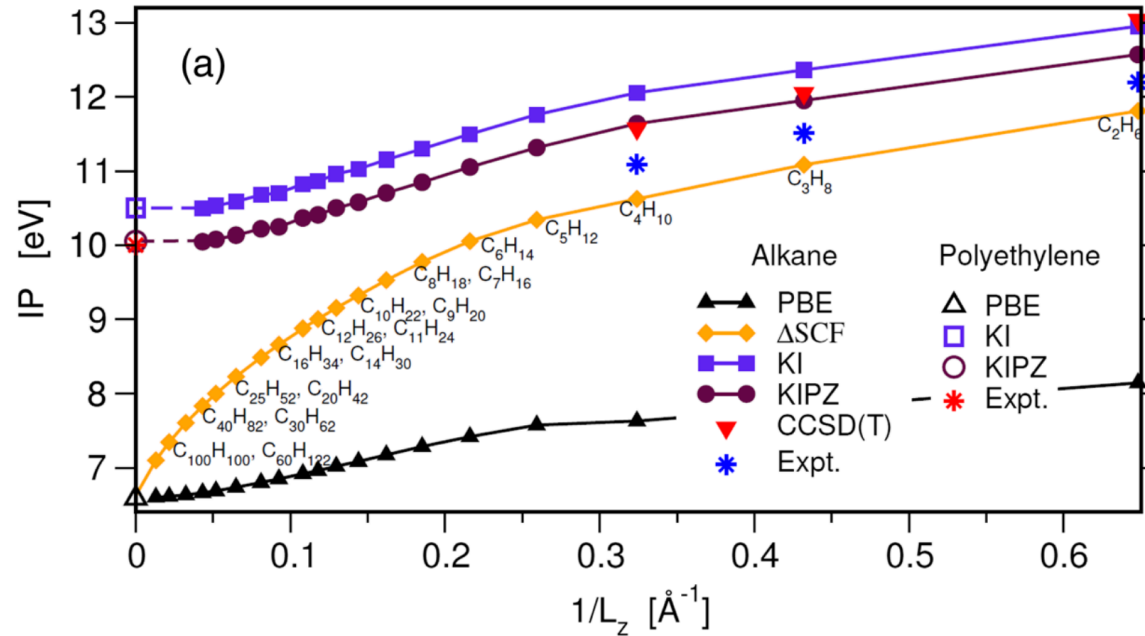
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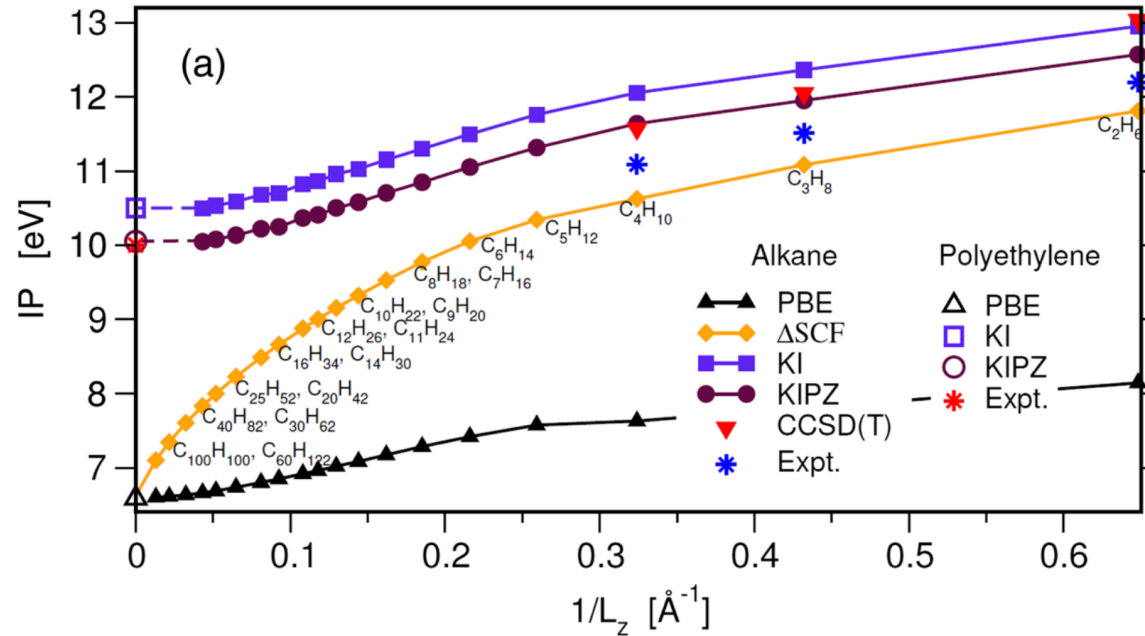
Across all the cells $\Delta E = \frac{1}{\delta N} (E(N - \delta N) - E(N)) = -\frac{dE}{dN} = -\varepsilon_{\text{HO}}$

Importance of localisation



$$\lim_{n_i(r) \rightarrow 0} v_i^{\text{KI}} / \alpha_i = \lim_{n_i(r) \rightarrow 0} -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}') = 0$$

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Solution: apply Koopmans correction to localised orbitals

Orbital-density-dependence



Other features of orbital-density-dependence

A. Ferretti *et al.* *Phys. Rev. B* **89**, 195134 (2014)

Orbital-density-dependence



Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general

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Orbital-density-dependence



Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general
- Practically we can often use MLWFs
- a natural generalisation in the direction of spectral functional theory

Screening

In **Hartree-Fock** (the original “Koopmans’ theorem”)

$$E_{\text{ee}}^{\text{HF}} = \frac{1}{2} \sum_{ij} f_i f_j \int d\mathbf{r} d\mathbf{r}' \frac{|\psi_i(\mathbf{r})|^2 |\psi_j(\mathbf{r}')|^2}{r - r'} - \frac{\psi_i^*(\mathbf{r}) \psi_j^*(\mathbf{r}') \psi_i(\mathbf{r}') \psi_j(\mathbf{r})}{r - r'}$$

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$$\frac{d^2 E_{\text{ee}}^{\text{HF}}}{df_i df_j} \stackrel{?}{=} 0$$

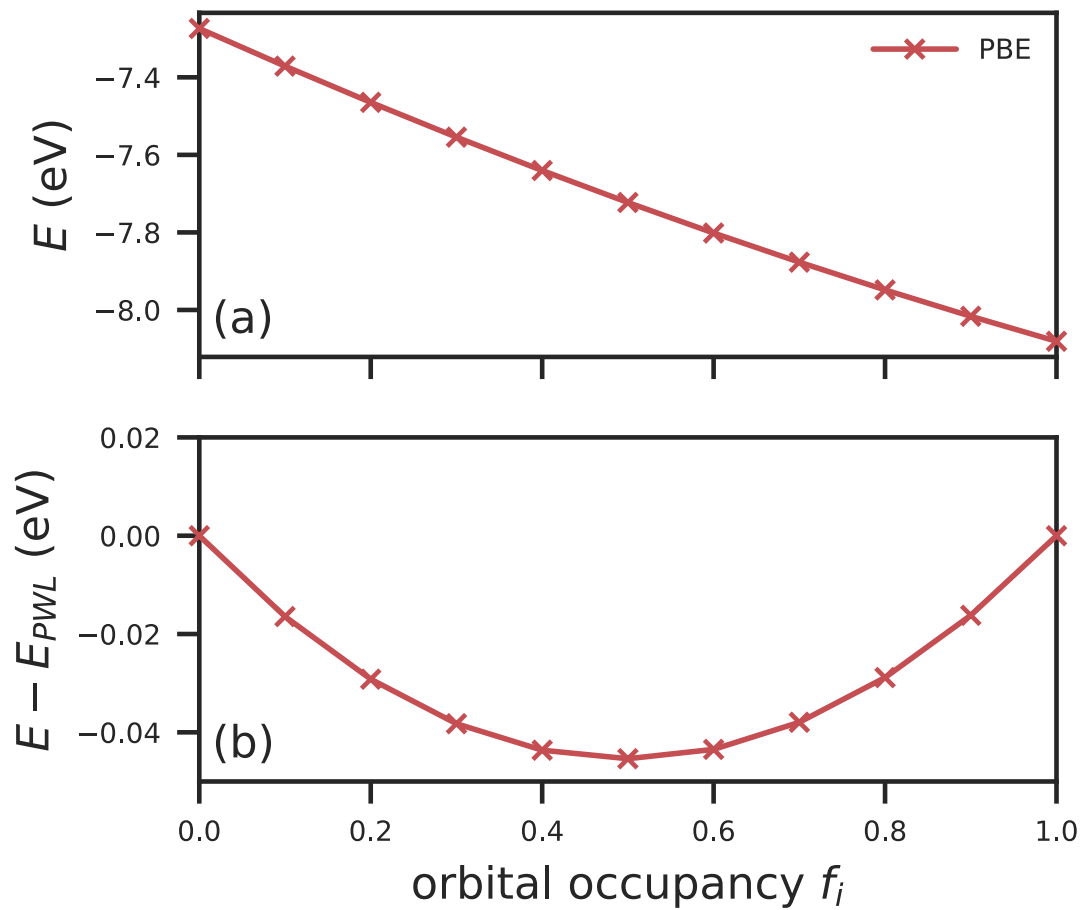
Koopmans functionals

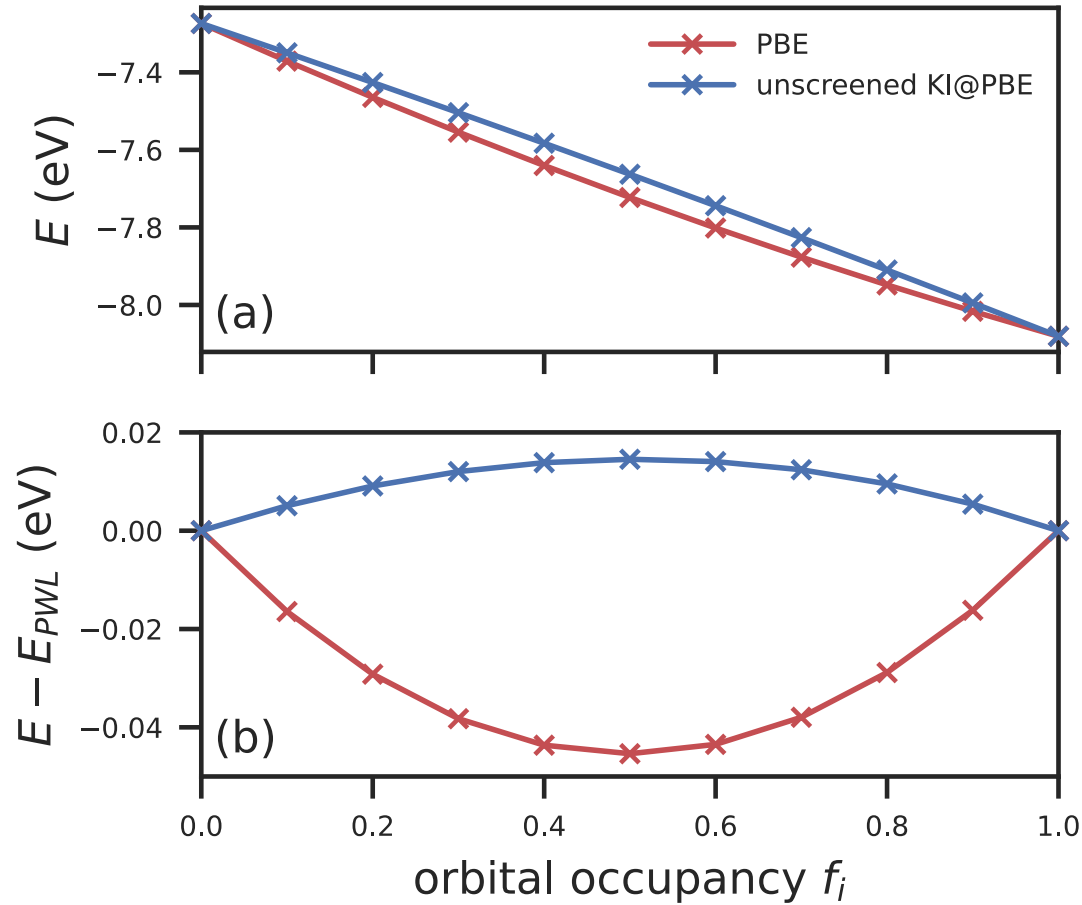
$$E_{\text{KI}}[\{\rho_i\}] = E_{\text{DFT}}[\rho] + \sum_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df \right)$$

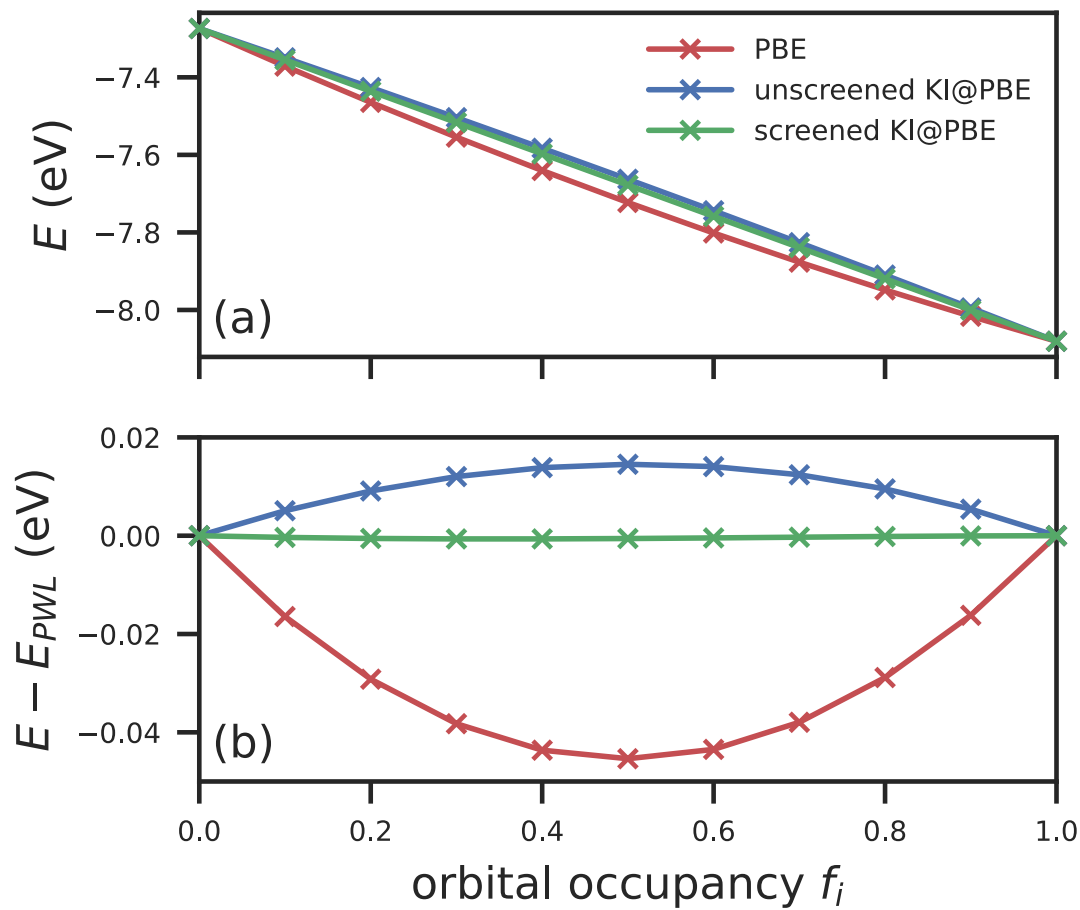
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$$\begin{aligned} E_{\text{KI}}[\{\rho_i\}] &= E_{\text{DFT}}[\rho] + \sum_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df \right) \\ &= E_{\text{DFT}}[\rho] + \sum_i \left(- \left(E_{\text{DFT}}[\rho] - E_{\text{DFT}}[\rho^{f_i \rightarrow 0}] \right) + f_i \left(E_{\text{DFT}}[\rho^{f_i \rightarrow 1}] - E_{\text{DFT}}[\rho^{f_i \rightarrow 0}] \right) \right) \stackrel{?}{=} \end{aligned}$$

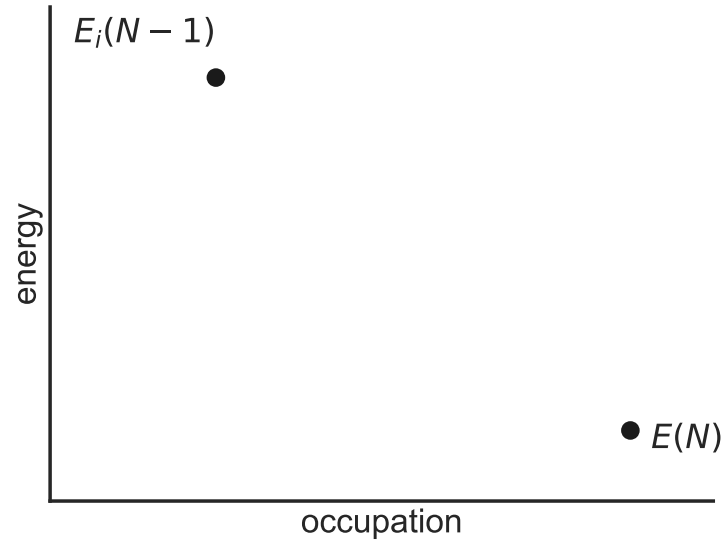
cannot be evaluated directly total energy differences



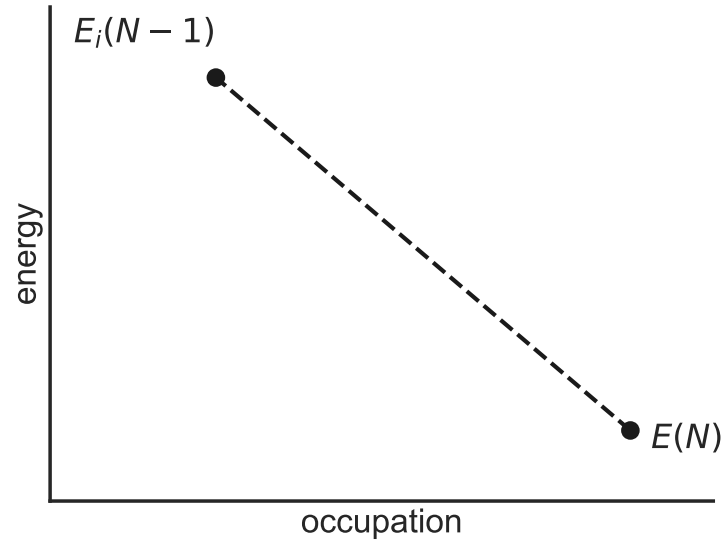




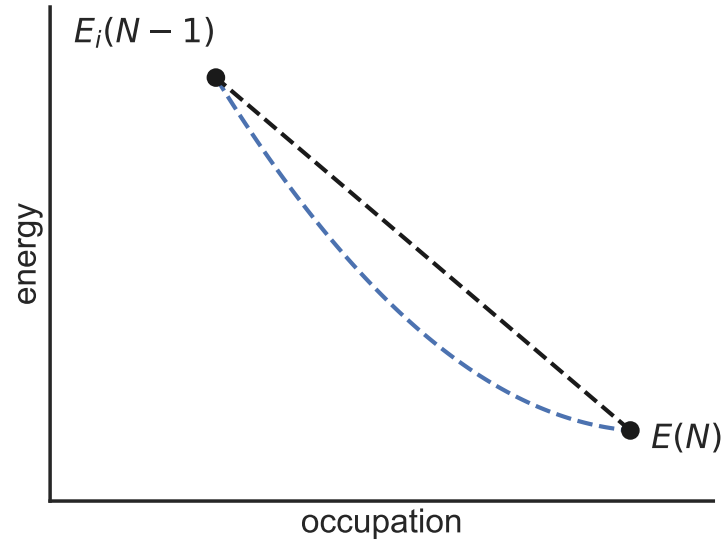
Screening: how to calculate the ideal α ?



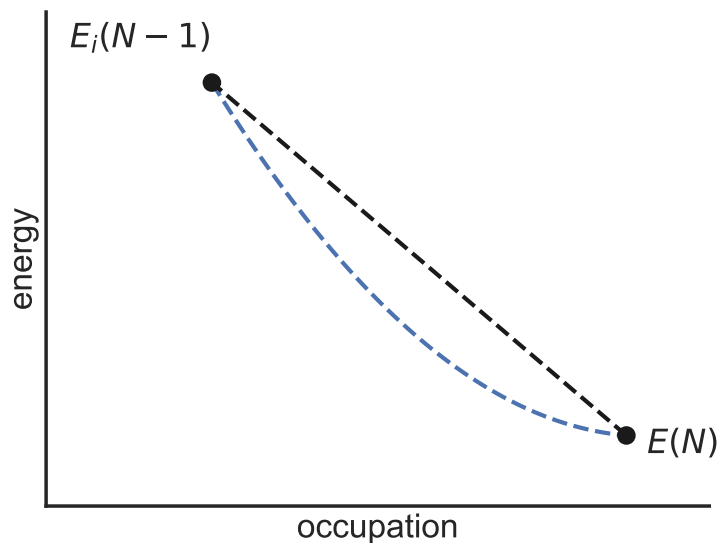
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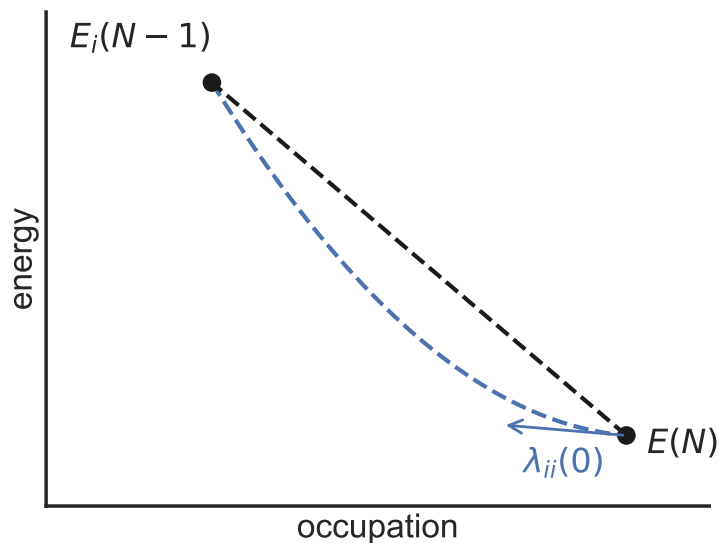


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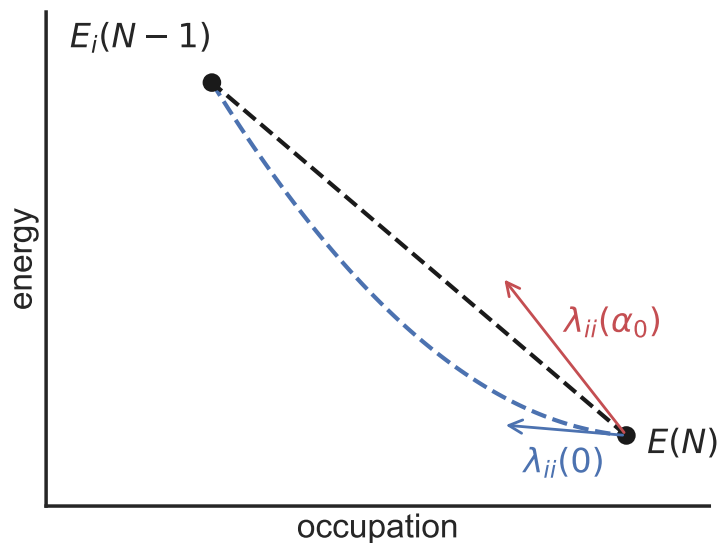
$$\lambda_{ii}(\alpha) \equiv \langle \varphi_i | \hat{h}^{\text{DFT}} + \alpha \hat{v}^{\text{Koopmans}} | \varphi_i \rangle = \left. \frac{dE^{\text{Koopmans}}}{df_i} \right|_{f_i=s}$$

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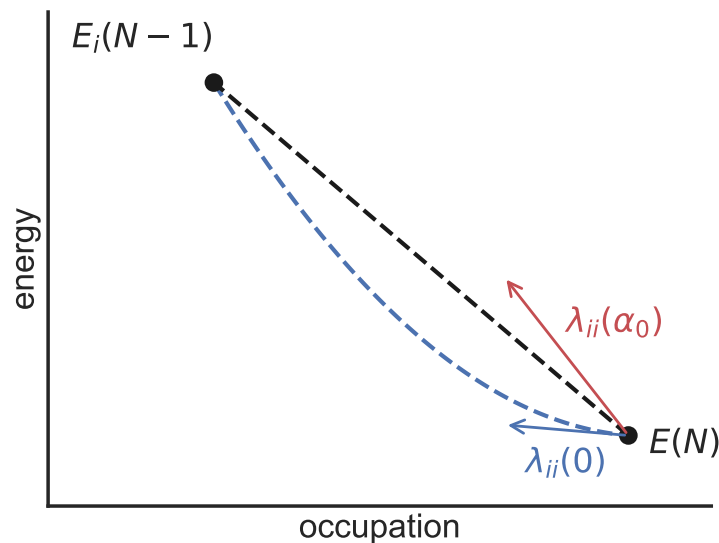
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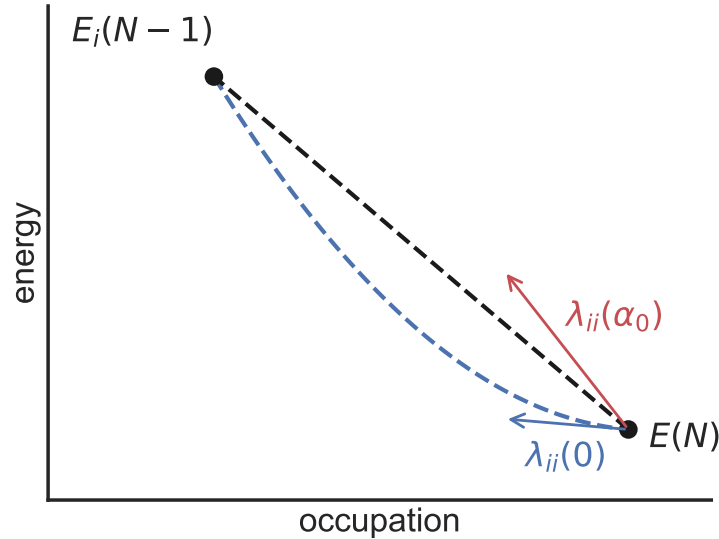
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Screening: how to calculate the ideal α ?



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$$\alpha_{n+1} = \alpha_n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha_n) - \lambda_{ii}(0)}$$

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

screening parameter



total energy with electron
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↓

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Diagram illustrating the screening parameter α and its iterative update:

- $E_i(N-1)$: total energy with electron removed from orbital i
- $E(N)$: total energy of neutral system
- $\lambda_{ii}(0)$: screening parameter at $\alpha=0$
- $\lambda_{ii}(\alpha^n)$: screening parameter at α^n

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Diagram illustrating the screening parameter α and its iterative update:

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- α^n : screening parameter

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Diagram illustrating the screening coefficient formula with annotations:

- $E_i(N-1)$: total energy with electron removed from orbital i
- $E(N)$: total energy of neutral system
- $\lambda_{ii}(0)$: expectation value of $\hat{H}^{\text{DFT}}$
- $\lambda_{ii}(\alpha^n)$: expectation value of $\hat{H}^{\text{Koopmans}}$
- α^n : screening parameter

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

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Reformulate in terms of DFPT...

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... in reciprocal space

$$\alpha_{\mathbf{0}i} = 1 + \frac{\sum_{\mathbf{q}} \langle v_{\text{pert},\mathbf{q}}^{\mathbf{0}i} \mid \Delta_{\mathbf{q}}^{\mathbf{0}i} n \rangle}{\sum_{\mathbf{q}} \langle n_{\mathbf{q}}^{\mathbf{0}i} \mid v_{\text{pert},\mathbf{q}}^{\mathbf{0}i} \rangle}$$

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N.B. even for the supercell, we can still reconstruct a band structure

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- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
 - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
 - are we going to explicitly minimise the ODD?

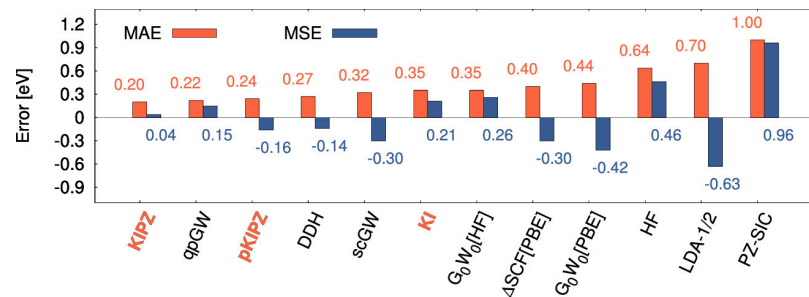
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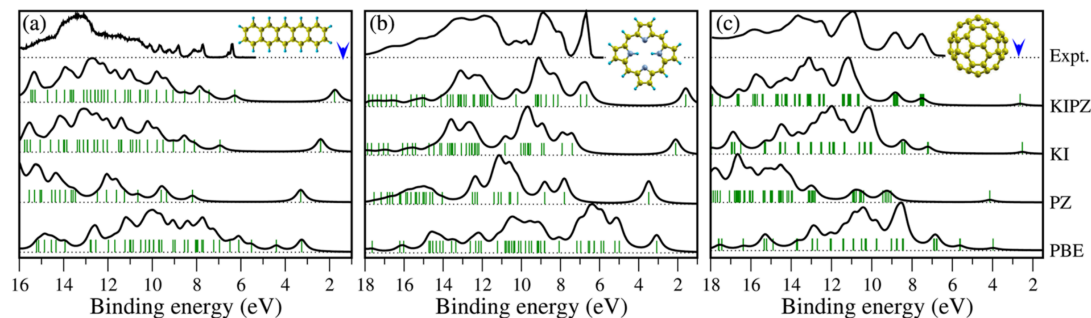
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 - are we going to explicitly minimise the ODD?
- how are we calculating the screening parameters? (finite differences, DFPT, ML...)

Results

Ionisation potentials of 100 molecules cf. CCSD(T)

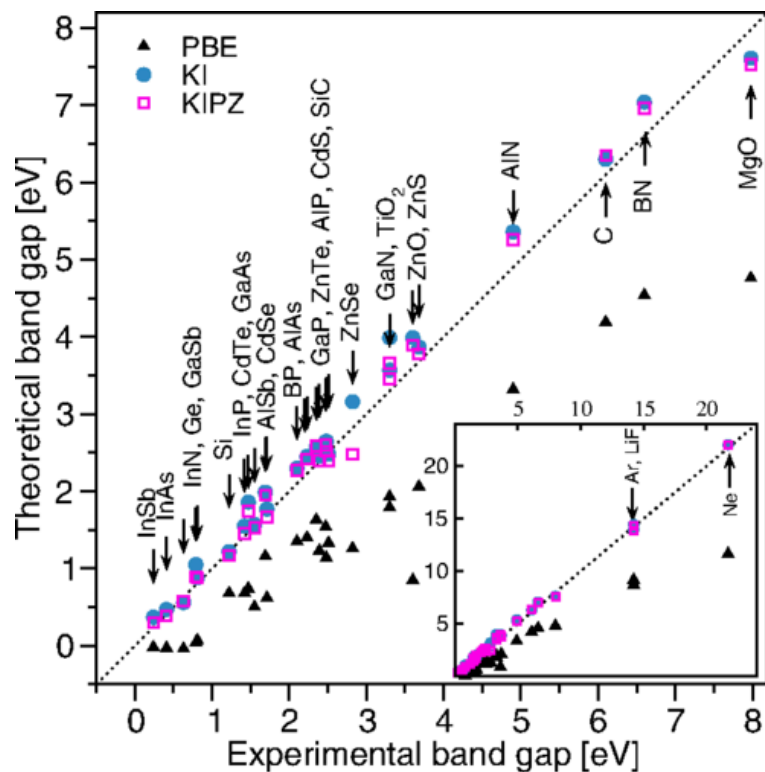


Ultraviolet photoemission spectra



N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018), N. L. Nguyen *et al.* *Phys. Rev. Lett.* **114**, 166405 (2015)

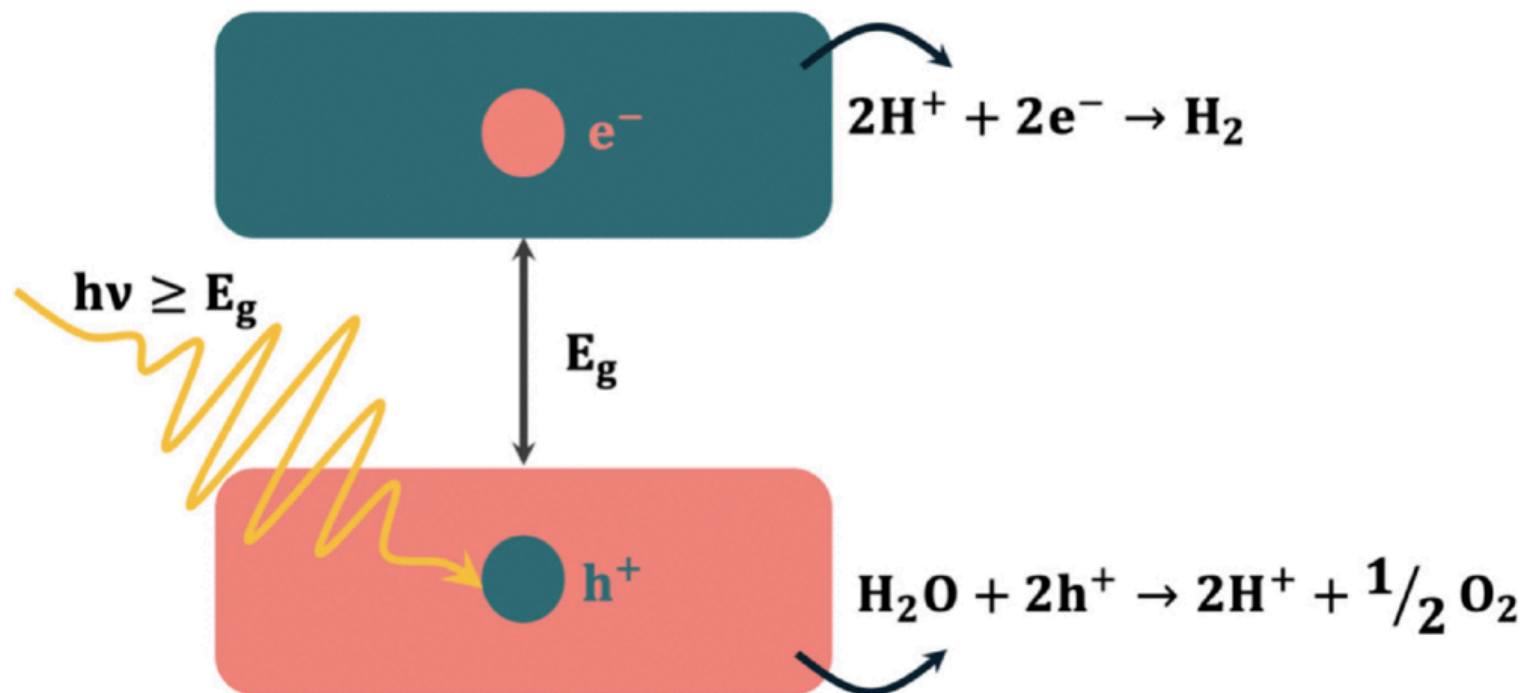
Koopmans functionals: results for solids



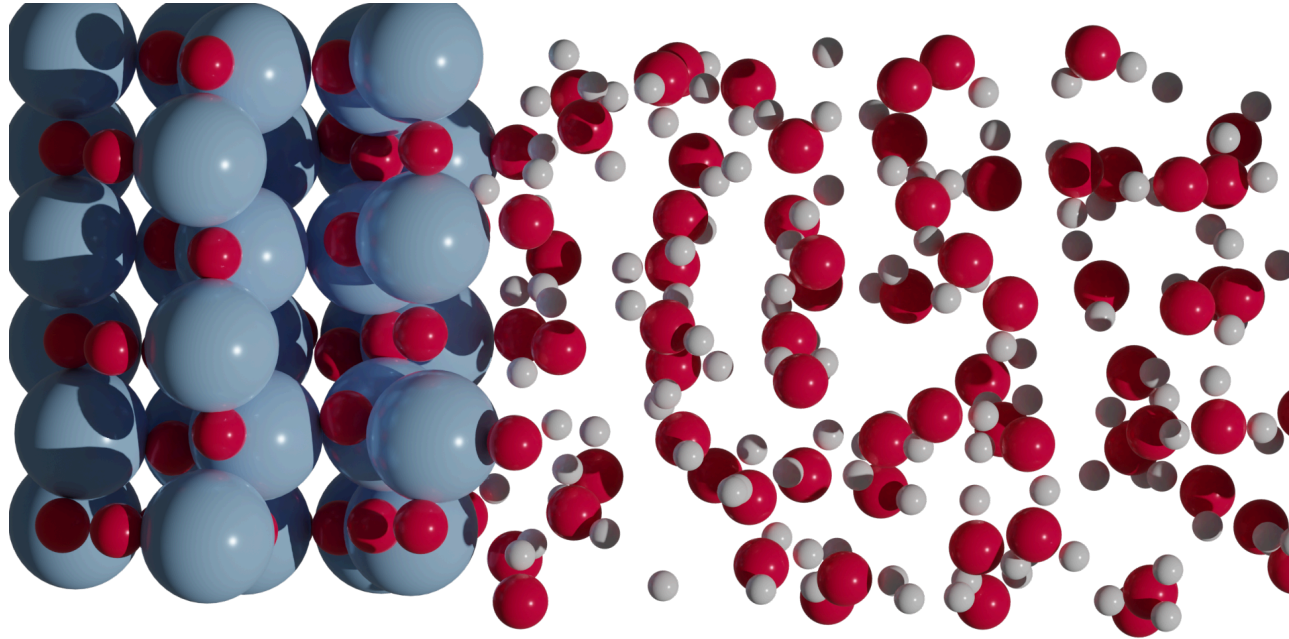
Mean absolute error (eV) across prototypical semiconductors and insulators

	PBE	G_0W_0	KI	KIPZ	QSGW $\tilde{W}$
E_{gap}	2.54	0.56	0.27	0.22	0.18
IP	1.09	0.39	0.19	0.21	0.49

Photocatalytic water-splitting

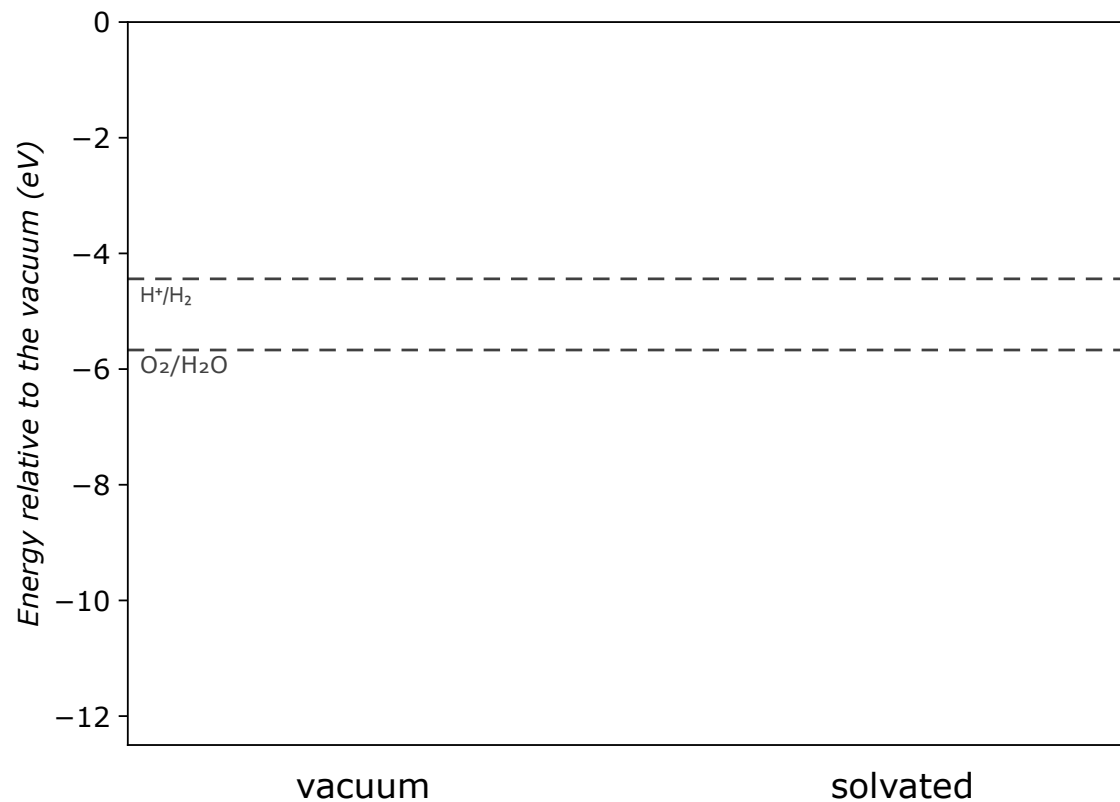


Photocatalytic water-splitting



M. Stojkovic *et al.* *Phys. Rev. Mater.* **10**, 25403 (2026)

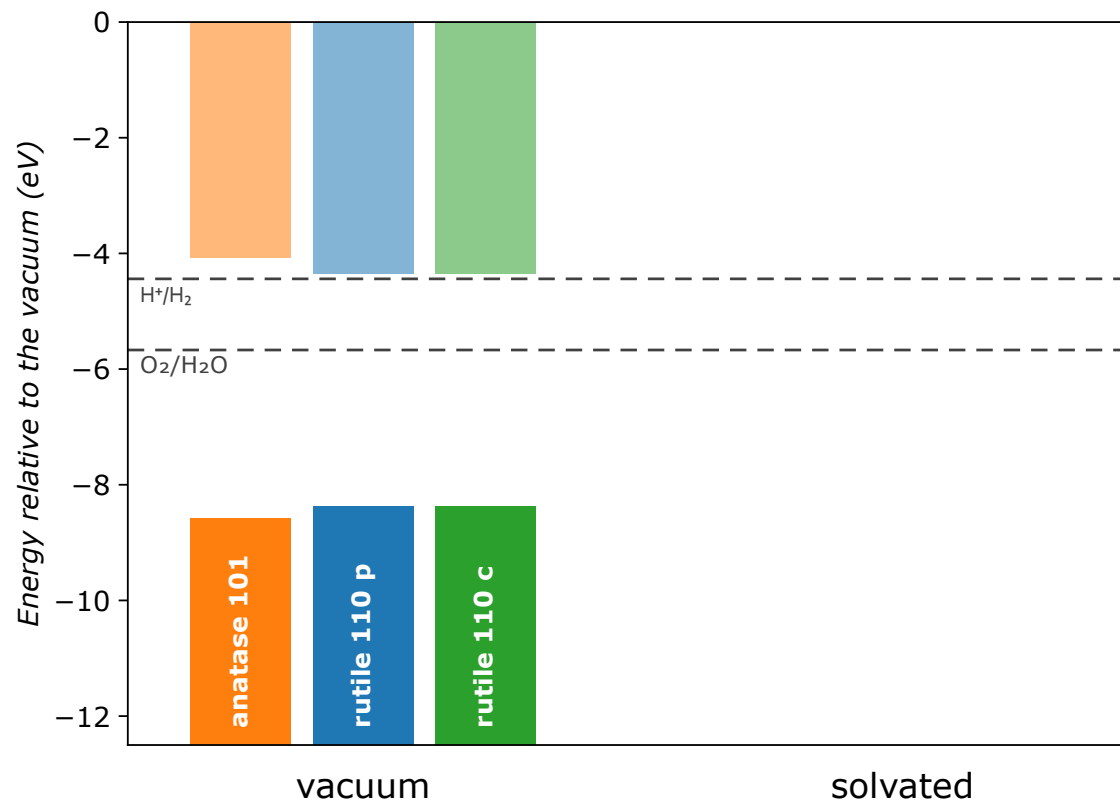
Band alignment for water-splitting



Band alignment for water-splitting

Slab calculations only semi-local!

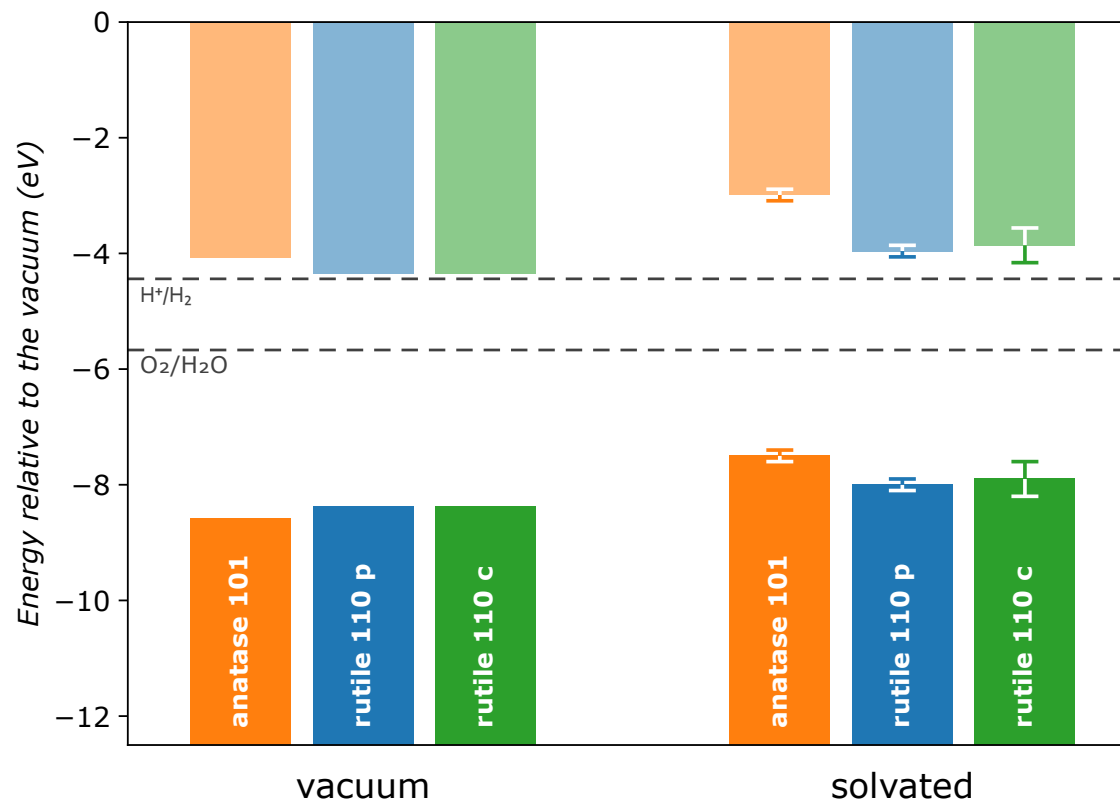
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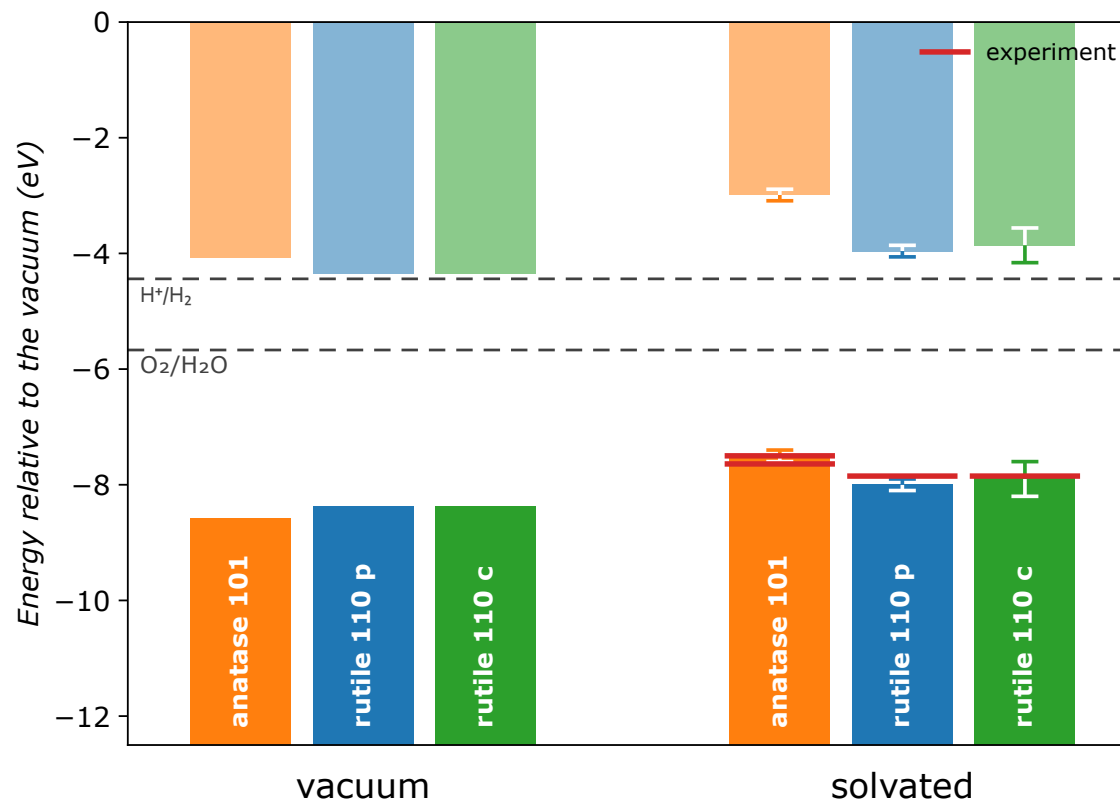
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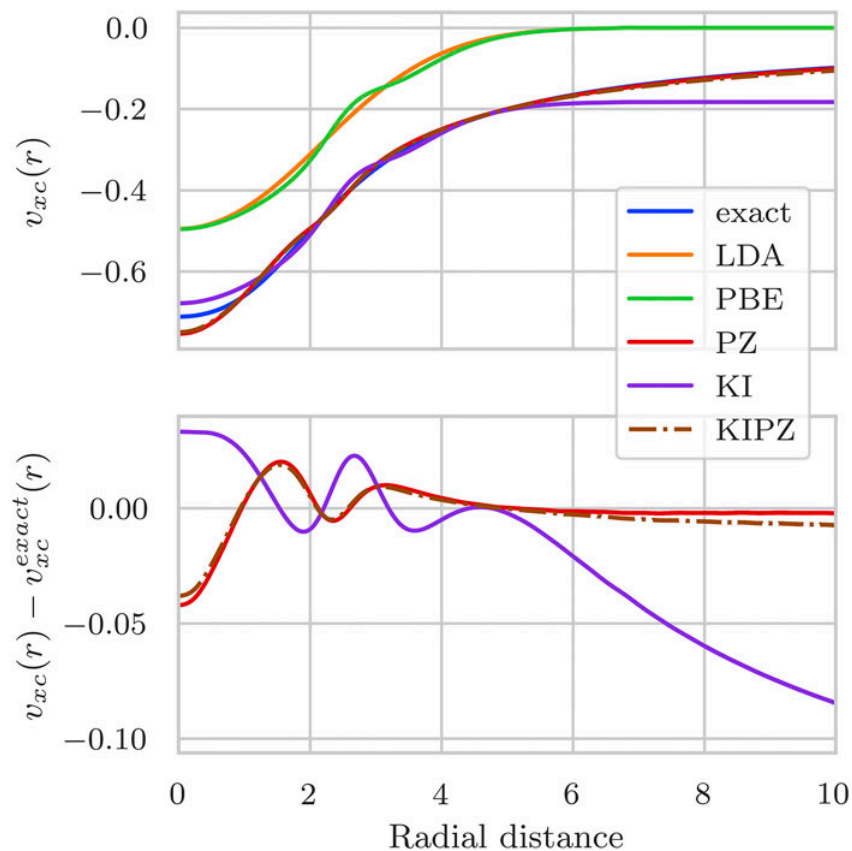


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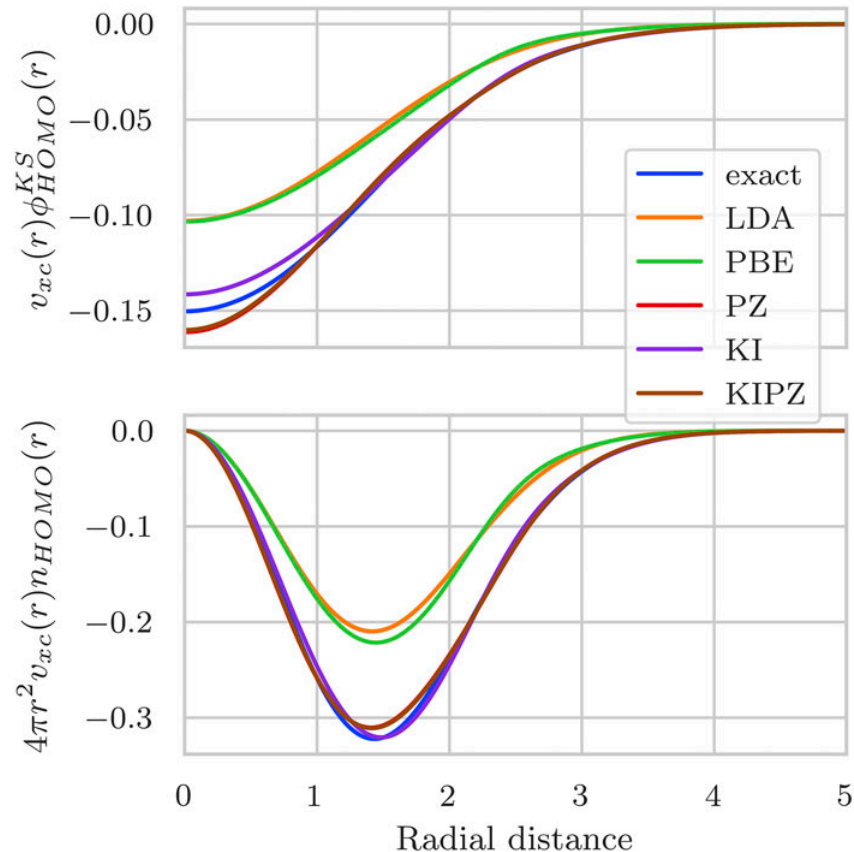
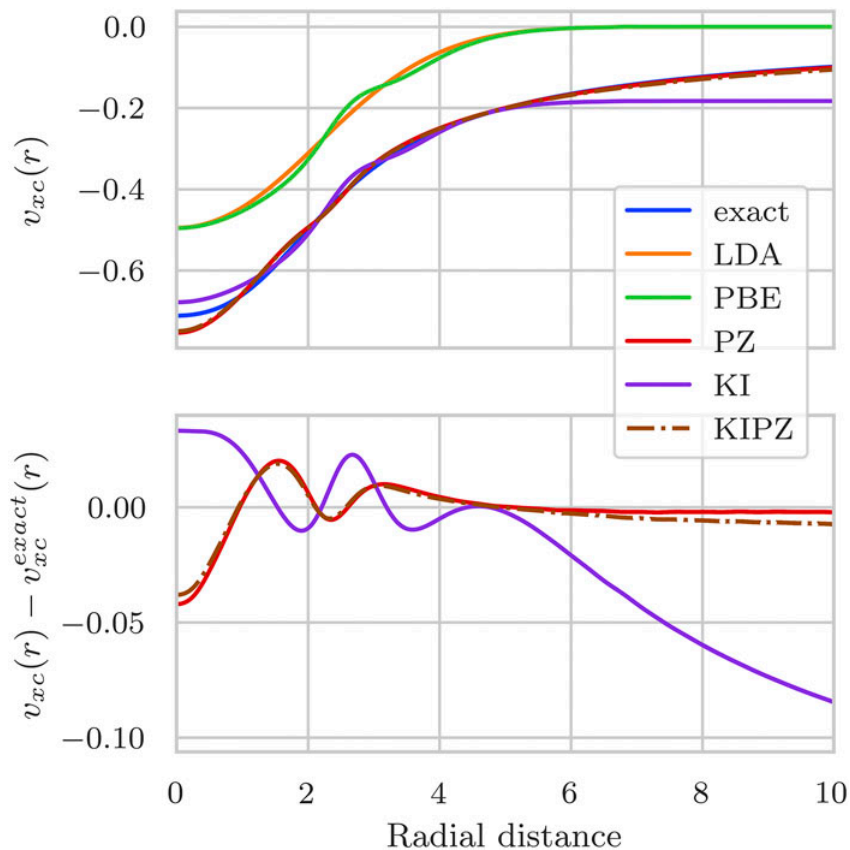
Toy systems

For Hooke's atom (two electrons in a harmonic confining potential with Coulombic repulsion)



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Koopmans functionals: caveats

N. Q. Su *et al.* *J. Phys. Chem. Lett.* **11**, 1528–1535 (2020)

Koopmans functionals: caveats

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- empty state localization in the bulk limit

Koopmans functionals: caveats



- restricted to systems with a non-zero band gap
- empty state localization in the bulk limit
- can potentially break the crystal point group symmetry

Resonance with other efforts

- Wannier transition-state method of Anisimov and Kozhevnikov
- Optimally tuned hybrid functionals of Kronik, Pasquarello, and others (refer back to Leeor's talk on Wednesday)
- Ensemble DFT of Kronik and co-workers
- Koopmans-Wannier of Wang and co-workers
- Dielectric-dependent hybrid functionals of Galli and co-workers
- LOSC functionals of Yang and co-workers

V. I. Anisimov *et al.* *Phys. Rev. B* **72**, 75125 (2005)

L. Kronik *et al.* *J. Chem. Theory Comput.* **8**, 1515–1531 (2012), D. Wing *et al.* *Proc. Natl. Acad. Sci.* **118**, e2104556118 (2021)

E. Kraisler *et al.* *Phys. Rev. Lett.* **110**, 126403 (2013)

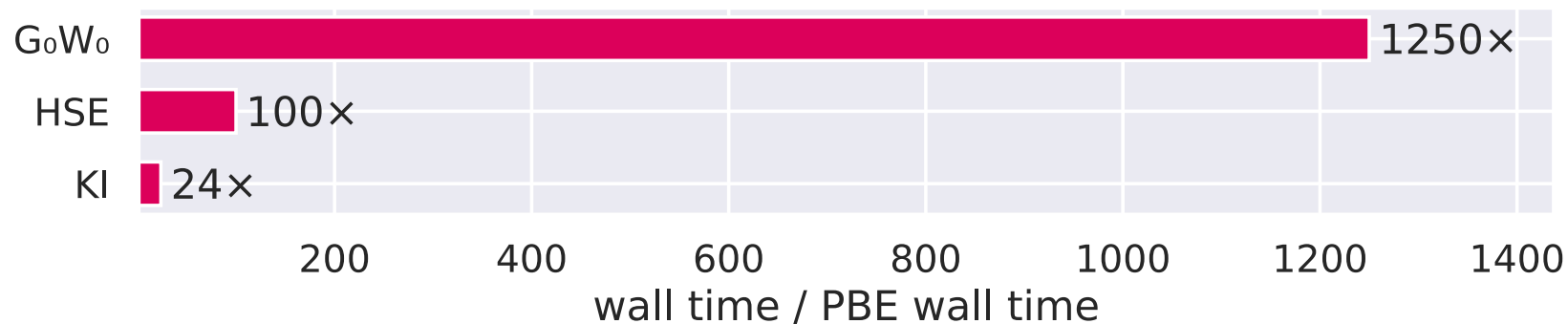
J. Ma *et al.* *Sci. Rep.* **6**, 24924 (2016)

J. H. Skone *et al.* *Phys. Rev. B* **93**, 235106 (2016)

C. Li *et al.* *Natl. Sci. Rev.* **5**, 203–215 (2018)

Computational cost and scaling

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Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\text{Hxc}} | n_i \rangle}{\langle n_i | f_{\text{Hxc}} | n_i \rangle}$$

N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022), N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018)

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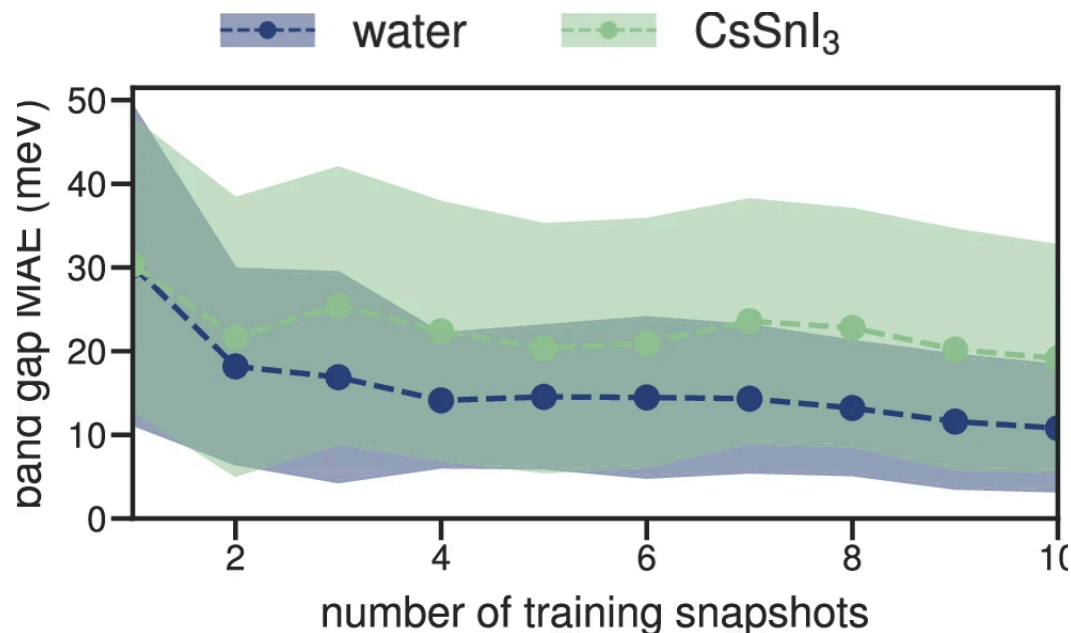
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 - DFPT: $\mathcal{O}(N_{\mathbf{k}}^2 N^3)$

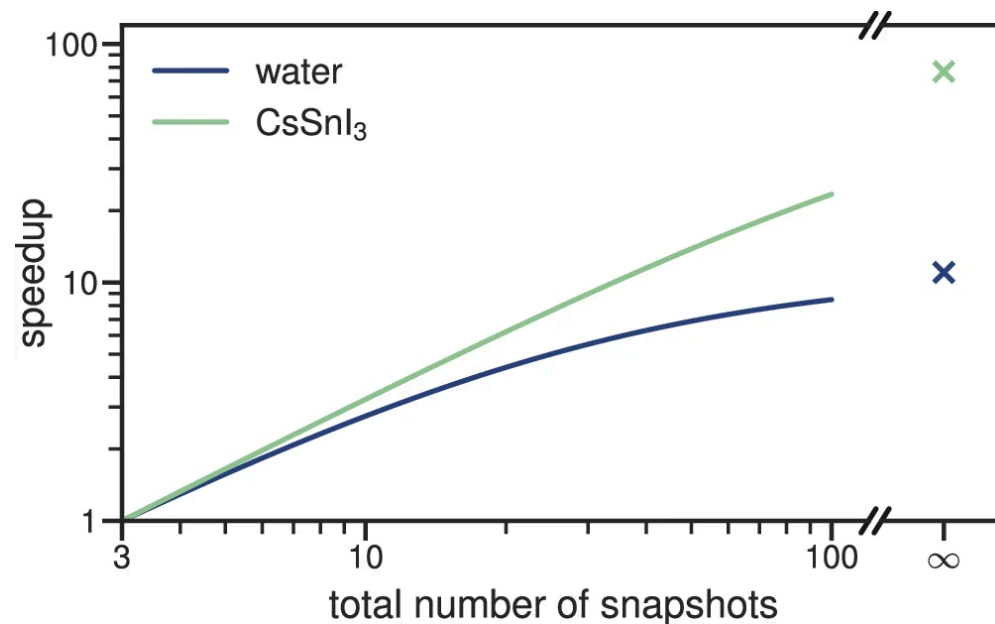
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Machine-learned electronic screening

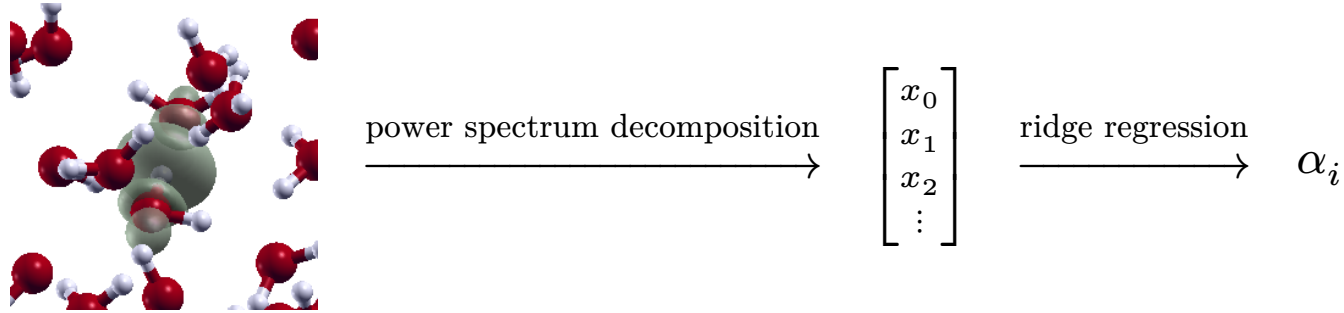


accurate to within $\mathcal{O}(10 \text{ meV})$ cf. typical band gap accuracy of $\mathcal{O}(100 \text{ meV})$



speedup of $\mathcal{O}(10)$ to $\mathcal{O}(100)$

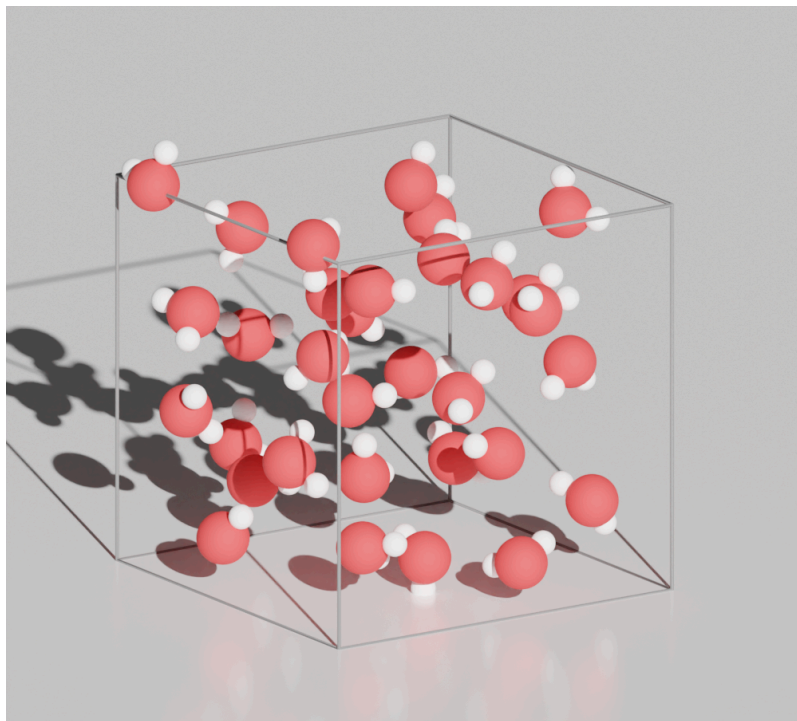
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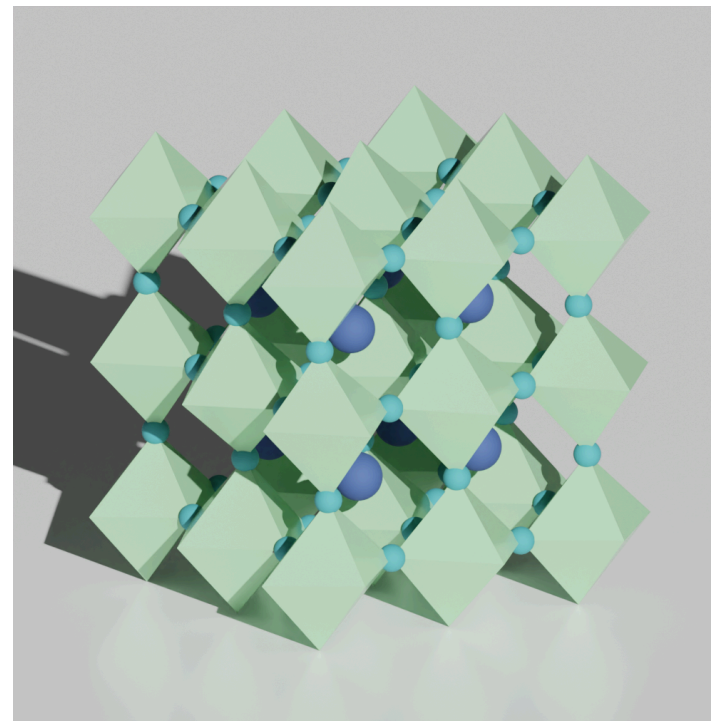
$$c_{nlm,k}^i = \int d\mathbf{r} g_{nl}(r) Y_{lm}(\theta, \varphi) n^i(\mathbf{r} - \mathbf{R}^i)$$

$$p_{n_1 n_2 l, k_1 k_2}^i = \pi \sqrt{\frac{8}{2l+1}} \sum_m c_{n_1 l m, k_1}^{i*} c_{n_2 l m, k_2}^i$$

Machine-learned electronic screening



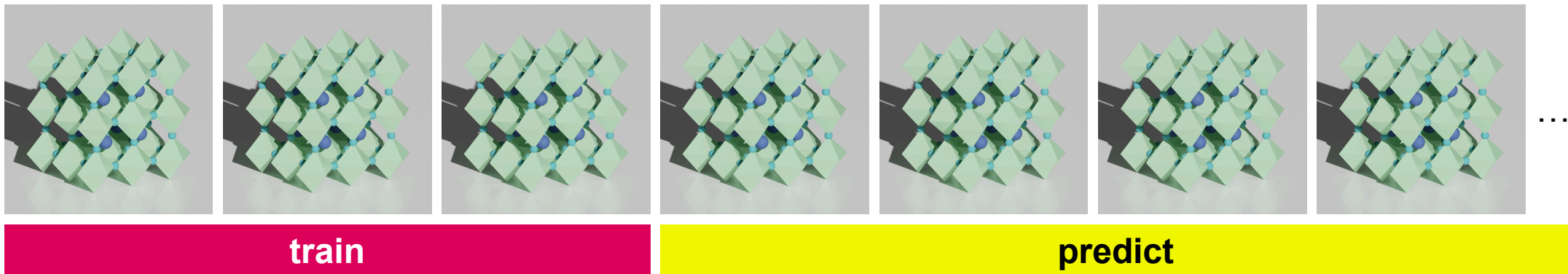
water



CsSnI_3

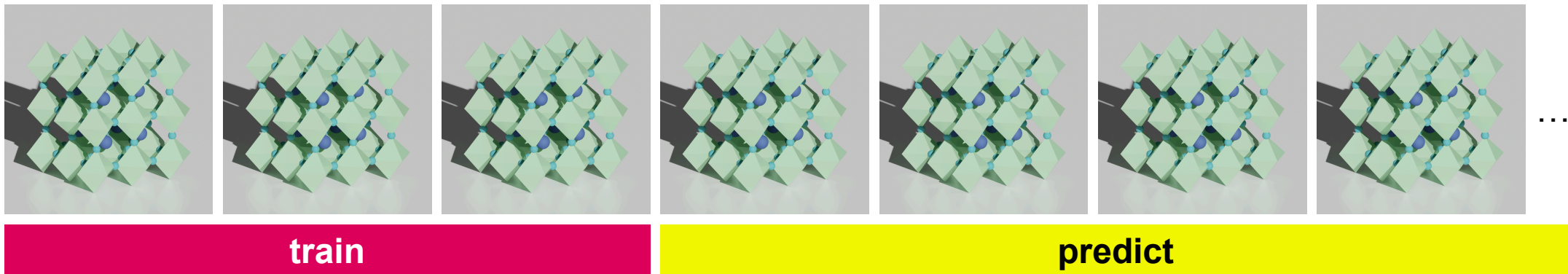
Machine-learned electronic screening

The use-case



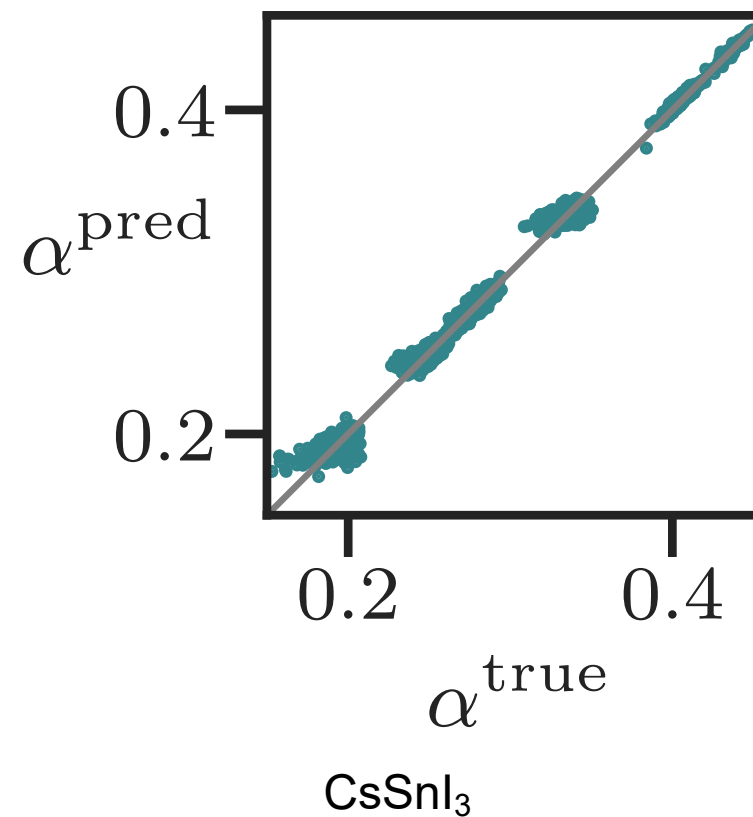
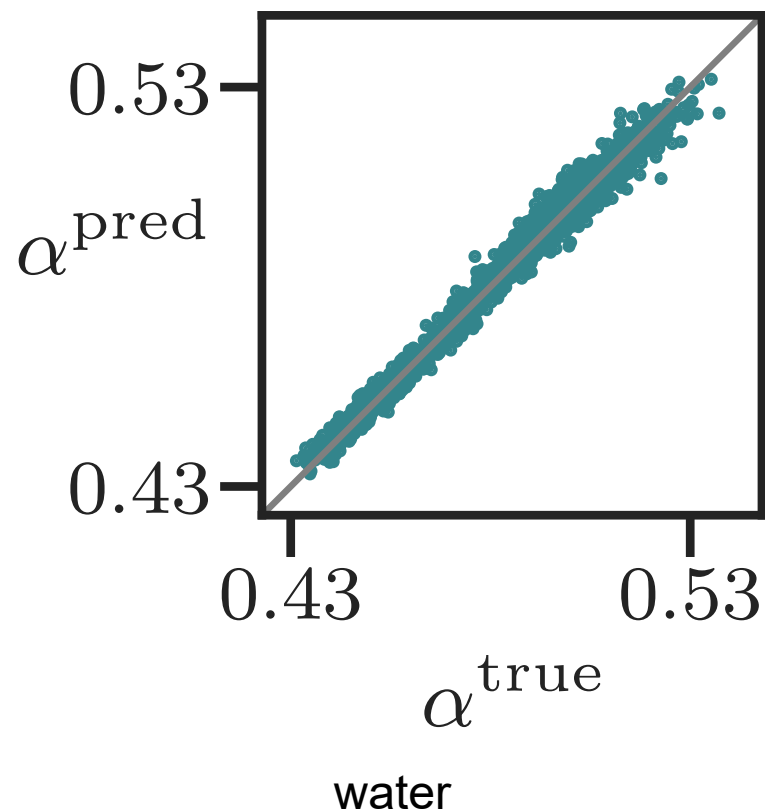
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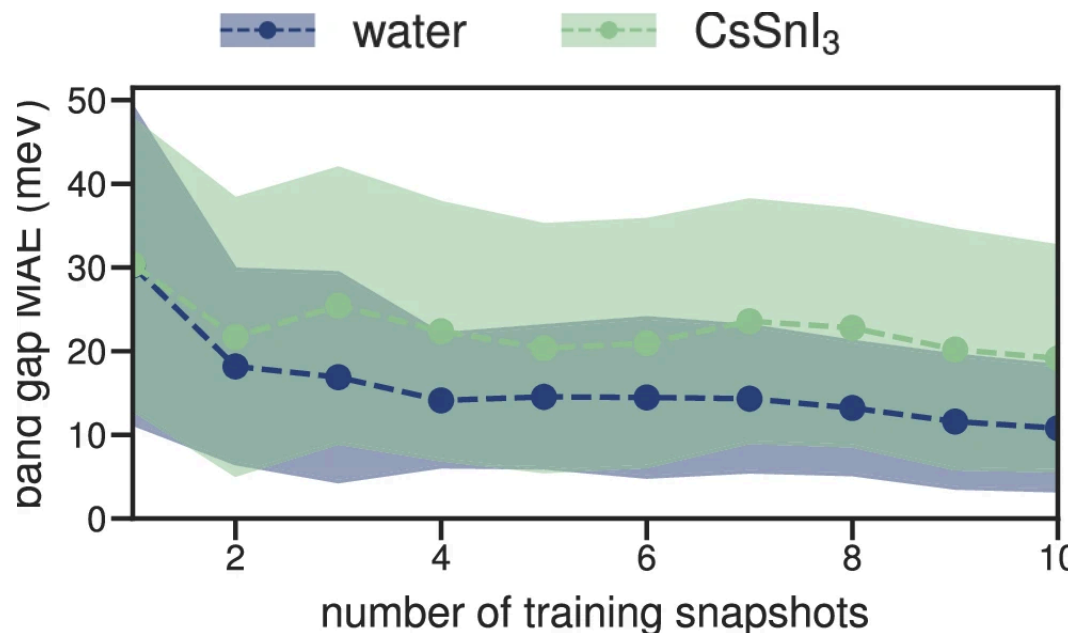


N.B. not a general model

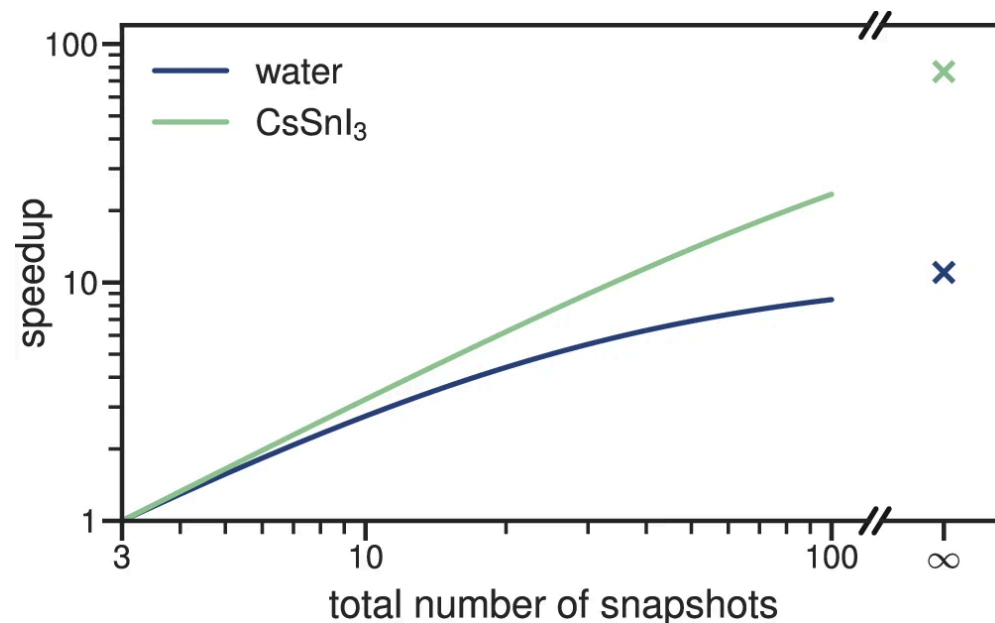
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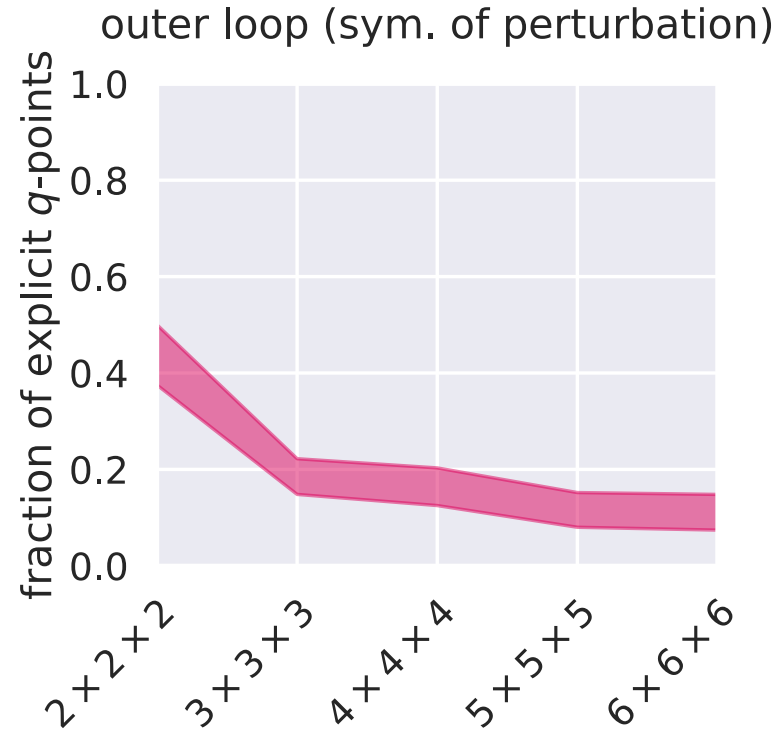
speedup of $\mathcal{O}(10)$ to $\mathcal{O}(100)$

Taking advantage of symmetries

To compute screening parameters via DFPT...

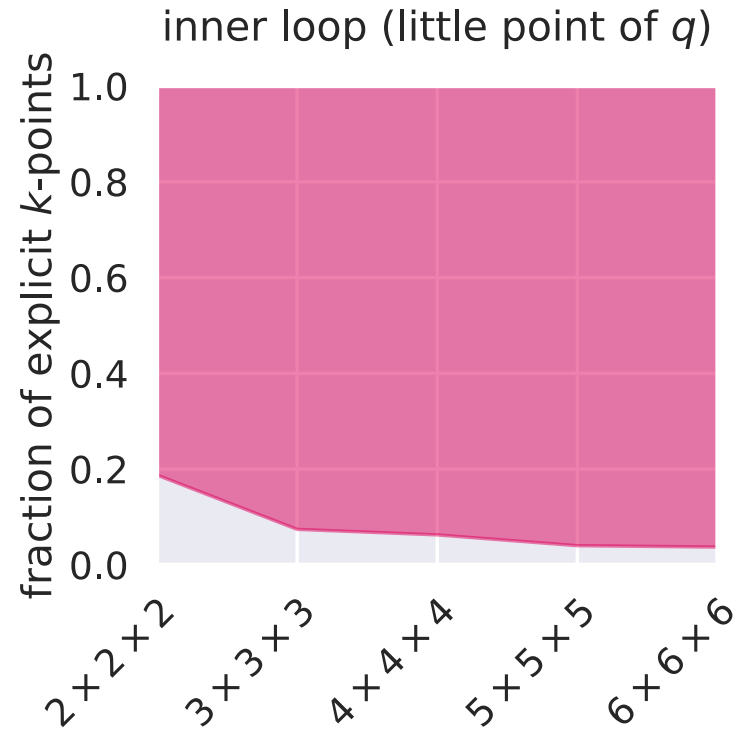
```
1: function CALCULATEALPHA( $n$ )
2:   for  $q \in \text{BZ}$  do
3:     for  $k \in \text{BZ}$  do
4:        $\triangleright$  Linear system  $Ax = b$  to obtain  $\Delta\psi_{\mathbf{k}+\mathbf{q},v}(\mathbf{r})$ 
5:     end
6:      $\Delta\rho_q^{0n} \leftarrow \sum_{\mathbf{k}v} \psi_{\mathbf{k}v}^*(\mathbf{r}) \Delta\psi_{\mathbf{k}+\mathbf{q},v}(\mathbf{r}) + c.c.$ 
7:      $\Pi_{0n,q}^{(r)} \leftarrow \langle \Delta\rho_q^{0n} | f_{\text{Hxc}} | \rho_q^{0n} \rangle$ 
8:      $\Pi_{0n,q}^{(u)} \leftarrow \langle \rho_q^{0n} | f_{\text{Hxc}} | \rho_q^{0n} \rangle$ 
9:   end
10:  return  $1 + \sum_q \Pi_{0n,q}^{(r)} / \sum_q \Pi_{0n,q}^{(u)}$ 
11:end
```

Taking advantage of symmetries



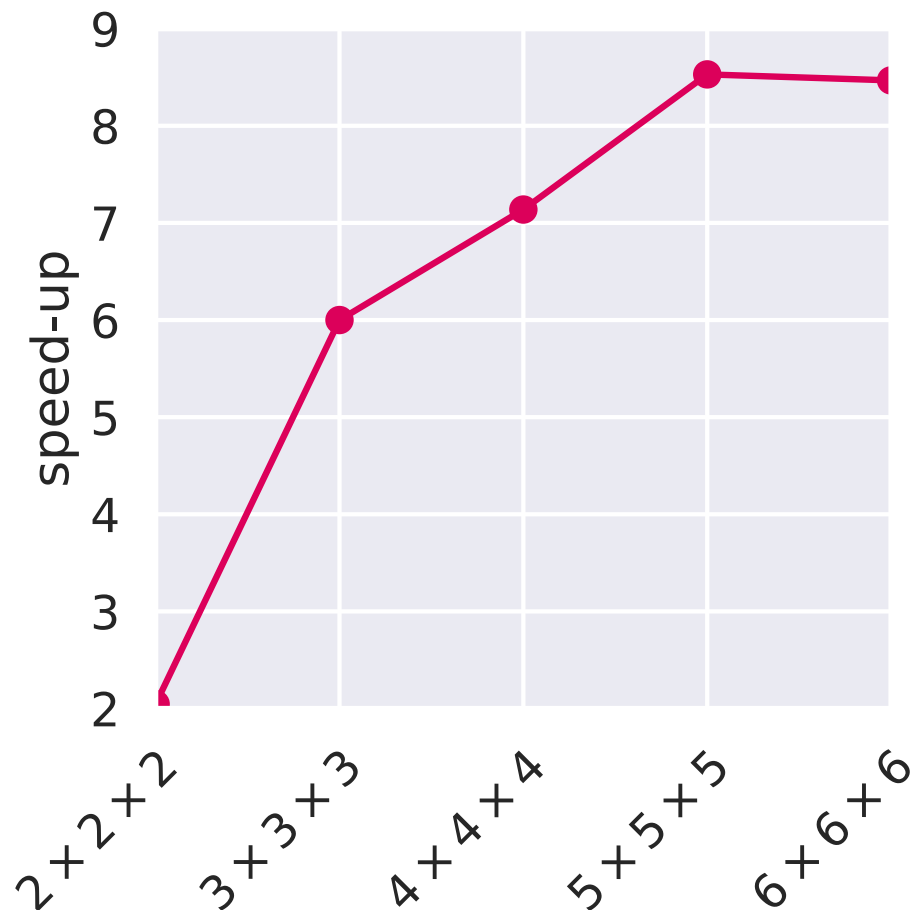
$\mathbf{q} \in \text{BZ} \rightarrow \mathbf{q} \in \text{IBZ}(n)$ (the symmetry of the perturbation; lower than that of the primitive cell)

Taking advantage of symmetries



$k \in \text{BZ} \rightarrow k \in \text{IBZ}(q)$ (can only use symmetries that leave q invariant)

Taking advantage of symmetries



Running a Koopmans functional calculation

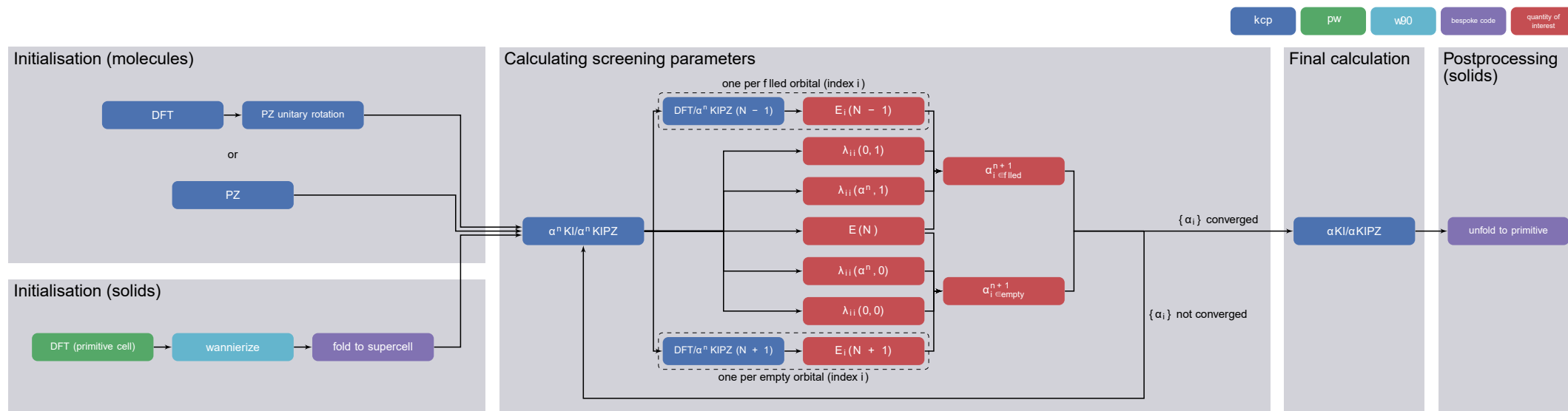
The workflows

The general workflow:

- define/initialize a set of variational orbitals
- calculate the screening parameters $\{\alpha_i\}$
- construct and diagonalize the Hamiltonian

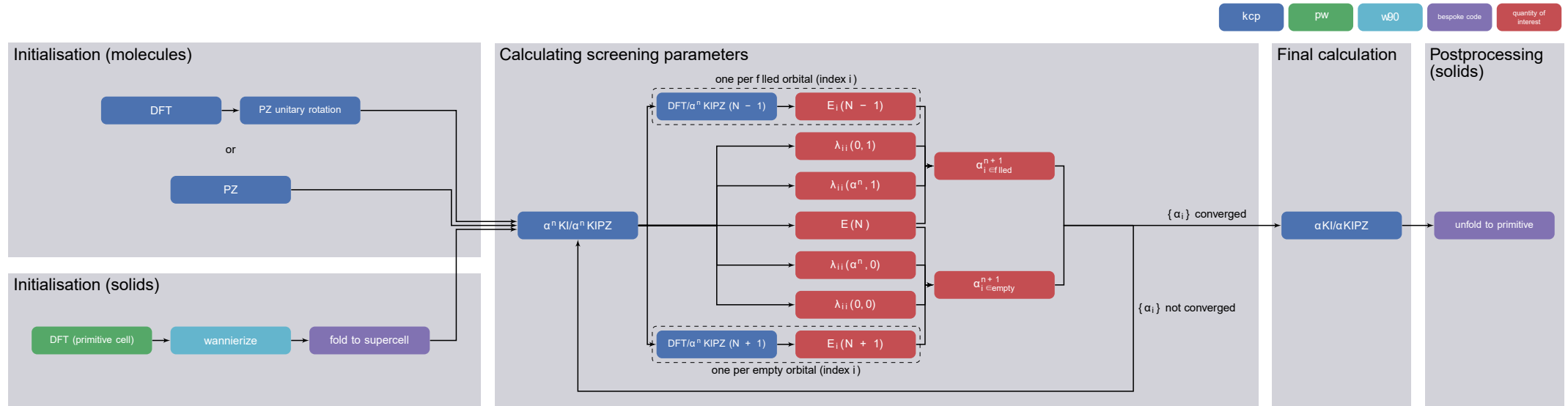
The workflows

(a) finite difference calculations using a supercell

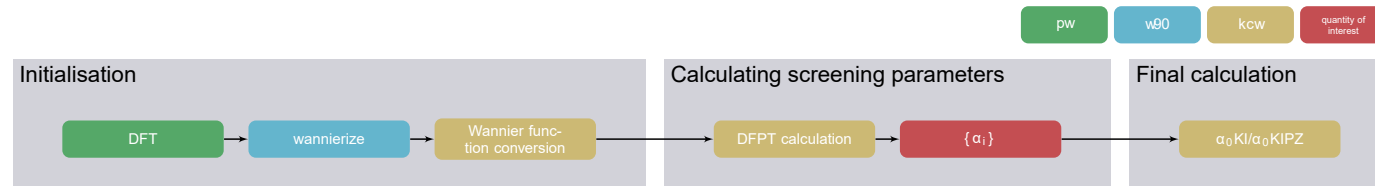


The workflows

(a) finite difference calculations using a supercell



(b) DFPT in a primitive cell



How do I run these calculations?

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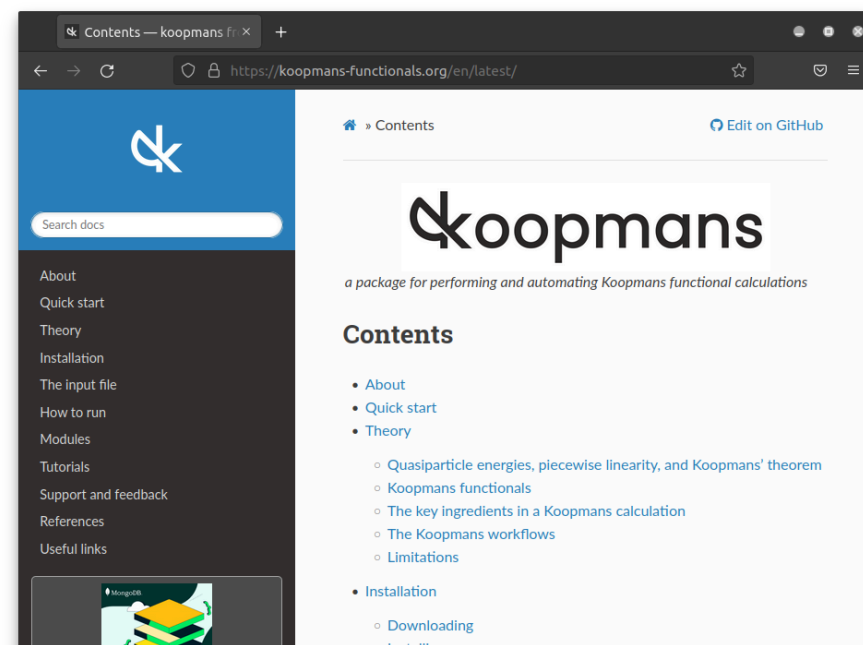
Our solution...

koopmans

The koopmans package

- v1.0 released in 2023 (now v1.2)
- implementations of Koopmans functionals
- automated workflows
 - start-to-finish Koopmans calculations
 - Wannierisation
 - dielectric tensor
 - ...
- built on top of ASE
- under the hood, calls QUANTUM ESPRESSO
- does not require expert knowledge

koopmans-functionals.org



E. B. Linscott *et al.* *J. Chem. Theory Comput.* **19**, 7097–7111 (2023)

A. H. Larsen *et al.* *J. Phys. Condens. Matter* **29**, 273002 (2017)

koopmans: the input file

```
{
  "workflow": {
    "task": "singlepoint",
    "functional": "ki",
    "method": "dscf",
    "init_orbitals": "mlwfs",
    "alpha_guess": 0.1
  },
  "atoms": {
    "atomic_positions": {
      "units": "crystal",
      "positions": [
        ["Si", 0.00, 0.00, 0.00],
        ["Si", 0.25, 0.25, 0.25]
      ]
    },
    "cell_parameters": {
      "periodic": true,
      "ibrav": 2,
      "celldm(1)": 10.262
    }
  },
}
```

```
  "k_points": {
    "grid": [8, 8, 8],
    "path": "LGXKG"
  },
  "calculator_parameters": {
    "ecutwfc": 60.0,
    "w90": {
      "projections": [
        [{"fsite": [0.125, 0.125, 0.125],
          "ang_mtm": "sp3"}],
        [{"fsite": [0.125, 0.125, 0.125],
          "ang_mtm": "sp3"}]
      ],
      "dis_froz_max": 11.5,
      "dis_win_max": 17.0
    }
  }
}
```

koopmans is scriptable



```
from ase.build import bulk
from koopmans.kpoints import Kpoints
from koopmans.projections import ProjectionBlocks
from koopmans.workflows import SinglepointWorkflow

# Use ASE to create bulk silicon
atoms = bulk('Si')

# Define the projections for the Wannierization (same for filled and empty manifold)
si_proj = [{'fsite': [0.25, 0.25, 0.25], 'ang_mtm': 'sp3'}]
si_projs = ProjectionBlocks.from_list([si_proj, si_proj], atoms=atoms)

# Create the workflow
workflow = SinglepointWorkflow(atoms = atoms,
                               projections = si_projs,
                               ecutfwc = 40.0,
                               kpoints = Kpoints(grid=[8, 8, 8], path='LGXKG', cell=atoms.cell),
                               calculator_parameters = {'pw': {'nbnd': 10},
                                                       'w90': {'dis_froz_max': 10.6, 'dis_win_max': 16.9}})

# Run the workflow
workflow.run()
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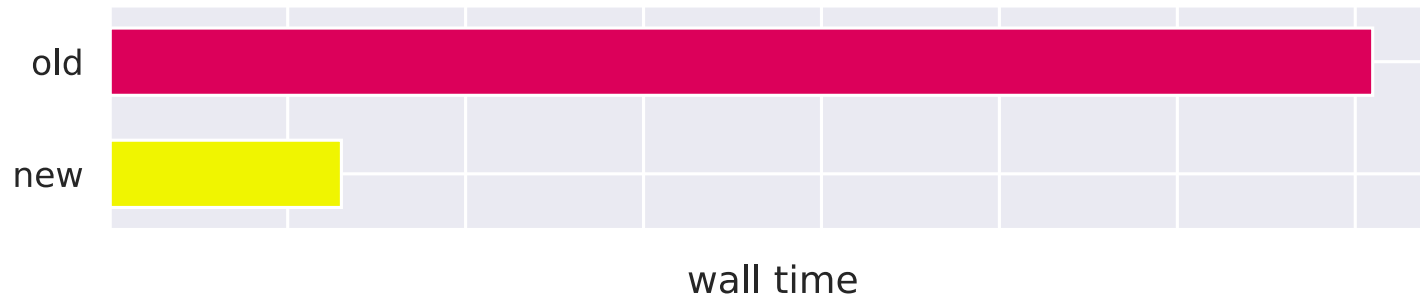
 **koopmans v2 is coming...** 

Koopmans AiiDA

- remote compute
- parallel step execution
- provenance-tracking
- error recovery
- and more!

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Automated Wannierisation

Koopmans functionals rely heavily on Wannier functions...

- to initialise the minimising orbitals, *or*
- in place of the minimising orbitals entirely

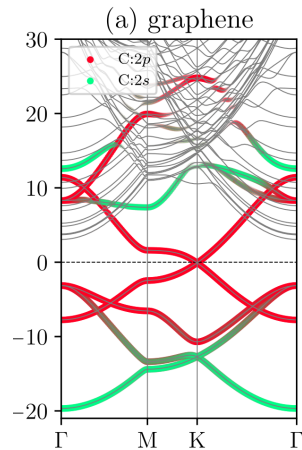
J. Qiao *et al. npj Comput Mater* **9**, 208 (2023)

J. Qiao *et al. npj Comput Mater* **9**, 206 (2023)

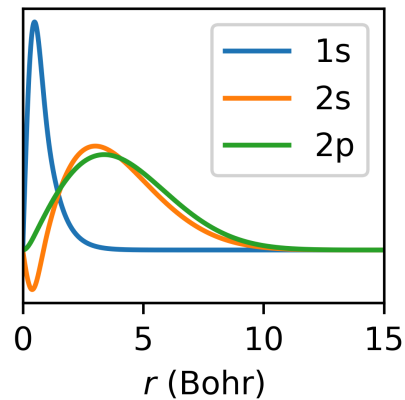
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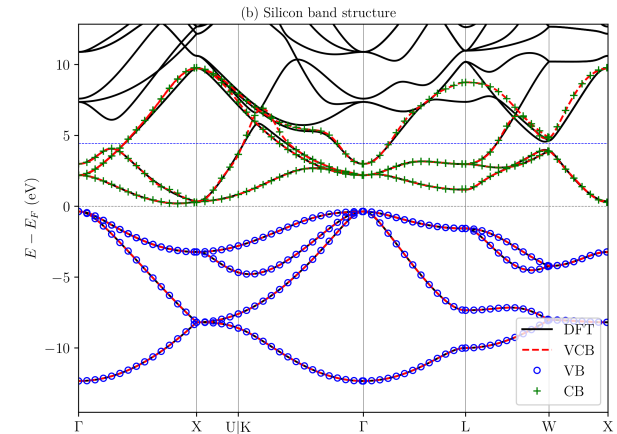
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projectability-based
disentanglement

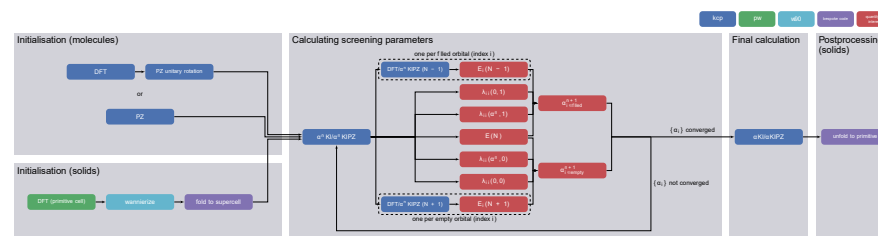
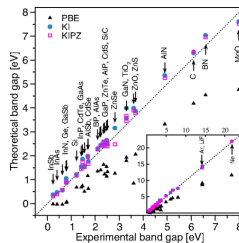
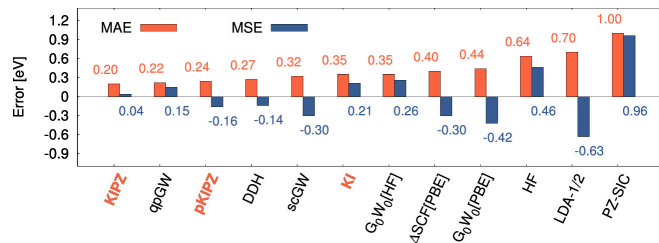


use PAOs found in
pseudopotentials



parallel transport to separate manifolds

Take home messages



- Koopmans functionals are more complicated than a simple semi-local DFT calculation, because of...
 - orbital-density-dependence
 - screening parameters
- Koopmans functionals are implemented in QUANTUM ESPRESSO
- the complexity of the workflows are handled by the koopmans package
- keep your eyes peeled for v2!

koopmans: An Open-Source Package for Accurately and Efficiently Predicting Spectral Properties with Koopmans Functionals

Edward B. Linscott,^{*,△} Nicola Colonna,[△] Riccardo De Gennaro, Ngoc Linh Nguyen, Giovanni Borghi, Andrea Ferretti, Ismaila Dabo, and Nicola Marzari^{*}



Cite This: <https://doi.org/10.1021/acs.jctc.3c00652>



Read Online

ACCESS |



Metrics & More

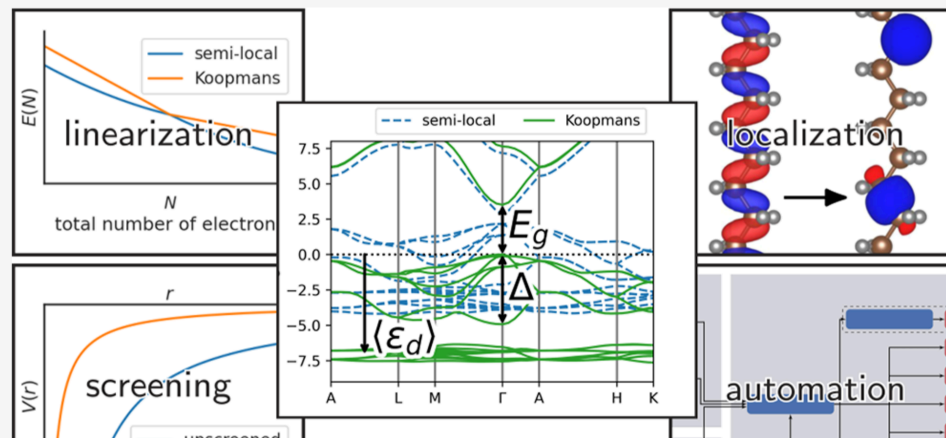


Article Recommendations



Supporting Information

ABSTRACT: Over the past decade we have developed Koopmans functionals, a computationally efficient approach for predicting spectral properties with an orbital-density-dependent functional framework. These functionals impose a generalized piecewise linearity condition to the entire electronic manifold, ensuring that orbital energies match the corresponding electron removal/addition energy differences (in contrast to semilocal DFT, where a mismatch between the two lies at the heart of the band gap problem and, more generally, the unreliability of Kohn–Sham orbital energies). This strategy has proven to be very powerful,



Thank you!

Acknowledgements



Nicola
Colonna



Miki
Bonacci



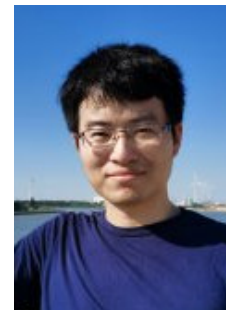
Aleksandr
Poliukhin



Marija
Stojkovic



Giovanni
Cistaro



Junfeng
Qiao



Yannick
Schubert



Nicola
Marzari



Want to find out more? Go to koopmans-functionals.org

slides available at  [github/elinscott-talks](https://github.com/elinscott-talks)

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Spare slides

Recap from earlier

Key idea: construct a functional such that the *variational* orbital energies

$$\varepsilon_i^{\text{Koopmans}} = \langle \varphi_i | H | \varphi_i \rangle = \partial E_{\text{Koopmans}} / \partial f_i$$

are...

- independent of the corresponding occupancies f_i
- equal to the corresponding total energy difference $E_i(N-1) - E(N)$

zero band gap $\rightarrow$ occupancy matrix for variational orbitals is off-diagonal